Nowcasting Growth-at-Risk using the Survey of Professional Forecasters*

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Abstract

This paper investigates nowcasting Growth-at-Risk (GaR) using consensus forecasts from the Survey of Professional Forecasters (SPF) in the US. Incorporating SPF consensus forecasts into the conditional mean of an AR-GARCH-type model significantly enhances nowcasting accuracy for GaR and the conditional density of GDP growth. Using US data from 1990 to 2024, the findings reveal strong time variation in both the lower and upper quantiles of the GDP growth distribution, underscoring the value of SPF consensus projections for real-time monitoring of tail risks to economic growth.

Keywords: Tail Risk, Macroeconomic Uncertainty, Density Forecasting, Real-Time Data
JEL Classification: C22, C52, C53

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1 Introduction

In monitoring the economic outlook, the focus of researchers and policymakers has recently expanded toward measuring economic downside risks. In particular, Growth-at-Risk (GaR), popularized by Adrian et al. (2019), typically represents the 5% or 10% quantile of the conditional GDP growth distribution as a measure of tail risk. This development has prompted major institutions, including the International Monetary Fund (IMF), and central banks like the Federal Reserve Bank of New York and the European Central Bank (ECB), to publish GaR forecasts for major economies.

So far, a substantial part of the literature employs Quantile Regressions (QR) to assess economic downside risk, indicating that lower quantiles of GDP growth vary with financial conditions (Giglio et al., 2016; Adrian et al., 2019, 2022; Ferrara et al., 2022). Conditional on financial indicators, the predictive GDP growth distribution exhibits time-varying lower quantiles, while the center and the upper part of the distribution are relatively constant over time. This has led to a surge in research findings that GDP growth is negatively skewed during economic downturns and vice versa (see, for example, Adrian et al., 2019; Delle Monache et al., 2024).

This paper proposes nowcasting quarterly GaR in real-time by leveraging the consensus forecasts from the US Survey of Professional Forecasters (SPF). Specifically, I use the median point predictions of the SPF as a predictor for the conditional mean of quarterly GDP growth. To assess the uncertainty around consensus forecasts, I employ a Generalized Autoregressive Conditional Heteroscedasticity (GARCH) model. The GARCH model, incorporating SPF consensus projections, provides timely nowcasts of GaR as well as the entire conditional density of GDP growth. A realistic out-of-sample evaluation demonstrates that incorporating the current economic outlook through SPF consensus forecasts significantly improves predictive performance for GaR and the conditional density of GDP growth. While the GARCH model exhibits slightly superior performance, QR with SPF forecasts also shows strong predictive accuracy.

In the context of predicting GaR, Brownlees and Souza (2021) provide evidence that a standard volatility model, such as the GARCH(1,1) model, tends to exhibit better out-of-sample predictive performance than the QR approach. Moreover, their findings suggest that the conditional GDP growth distribution is not significantly skewed and is moderately fat-tailed. Similarly, Carriero et al. (2022), including a large set of macroeconomic indicators in quantile regression models, also find that skewness is not a robust feature of GDP growth. Overall, these results highlight that both asymmetry and time-variation in higher moments crucially depend on the choice of the model and the conditioning information.

What these models have in common is that the practical application for real-time economic monitoring faces two main challenges. The first challenge stems from the delayed

releases of macroeconomic indicators, often referred to as the *ragged-edge* problem (Wallis, 1986). For instance, the first estimate of US GDP is typically released at the end of the first month of the subsequent quarter, while the nowcast from the US SPF becomes available mid-quarter. Additionally, macroeconomic indicators are available at different frequencies. While GDP is measured quarterly, some indicators like the Chicago Fed's National Financial Conditions Index (NFCI) are reported weekly, and others are even reported daily. This discrepancy in reporting frequencies necessitates the use of models capable of handling mixed-frequency data (see, for example, Ghysels et al., 2004, 2007; Giannone et al., 2008; Andreou et al., 2013). In the context of GaR, the timely incorporation of new data releases and revisions is crucial. Carriero et al. (2022) demonstrate that exploiting a large set of macroeconomic indicators in real-time improves the nowcast accuracy throughout the current quarter. Similarly, Ferrara et al. (2022) utilize daily financial conditions indicators in real-time for daily GaR predictions.

To circumvent the problem of delayed releases and mixed-frequency data, I exploit the consensus point predictions of the SPF. Rather than relying on a large set of macroeconomic indicators with varying release frequencies, the SPF offers a comprehensive assessment about the current economic outlook. The SPF is a quarterly survey that asks a panel of professional forecasters to provide point predictions of quarterly GDP for the current quarter and up to four quarters ahead (Clements et al., 2023). In particular, at short horizons of up to one quarter ahead, the SPF consensus point forecasts often outperform standard time-series models (Stark, 2010; Faust and Wright, 2013). This superior performance can likely be attributed to the aggregation of forecasts from multiple experts who efficiently incorporate a large pool of information and quickly adapt to major changes in the economic environment (Ang et al., 2007).

Regarding the prediction of conditional quantiles of GDP growth, the SPF is relatively unexplored. Adams et al. (2021) use financial conditions to explain fluctuations in the uncertainty around SPF consensus forecasts. Based on the conditional forecast error distribution, they construct GaR nowcasts, finding that incorporating financial conditions leads to significant out-of-sample improvements compared to GaR forecasts constructed solely from past forecast errors. Krüger et al. (2017) propose an approach based on entropic tilting to combine model-based and survey-based forecasts, offering a flexible way to incorporate SPF information into predictive distributions. The SPF also elicits density forecasts, which have been studied in the context of Growth-at-Risk by Ganics et al. (2024) and of Inflation-at-Risk by Andrade et al. (2014).

This paper contributes to the literature in several ways. First, it bridges the gap between the existing evidence on the short-horizon accuracy of the SPF consensus forecasts and the predictive performance of the GARCH model for forecasting GaR. By integrating

the SPF consensus forecasts into the GARCH model, this paper provides a tool for timely assessments of economic downside risks. Second, it relates to the nowcasting literature by leveraging the SPF in real-time, utilizing the timely information captured by the SPF consensus forecasts. This approach incorporates the current economic outlook without necessitating the inclusion of potentially numerous macroeconomic indicators with varying frequencies, delayed releases, and later revisions. Thirdly, this paper contributes to the literature on assessing the uncertainty around consensus forecasts. Instead of eliciting entire probability distributions, this paper constructs conditional densities by exploiting the forecast errors of the survey consensus forecasts (Reifschneider and Tulip, 2019; Clark et al., 2020; Adams et al., 2021).

The real-time out-of-sample evaluation reveals that the consensus forecasts of the SPF offer valuable information for GaR prediction. When conditioned on the SPF's median nowcasts, the GARCH model significantly outperforms quantile regressions with NFCI in predicting lower quantiles of the GDP growth distribution. Moreover, the NFCI provides limited additional predictive gains once the SPF nowcasts become available mid-quarter. Furthermore, the SPF's consensus projections are not only informative about downside risk but also about the upper quantiles of the GDP growth distribution. By accounting for changes in the conditional mean of the GARCH model, the SPF enhances forecasts of the entire GDP growth distribution. In addition, the out-of-sample analysis indicates that incorporating skewness and fat tails does not lead to an improved forecasting performance. While this observation does not necessarily imply symmetry in the true conditional distribution of GDP growth, it suggests that employing a more flexible distribution than the normal distribution lacks justification from a short-horizon forecasting perspective. Finally, the GARCH model with SPF projections in the conditional mean generates GaR nowcasts that perform on par with data-rich methods examined by Carriero et al. (2022).

The remainder of the paper is organized as follows. Section 2 provides a review of the related literature. Section 3 details the data and the associated release calendar. The forecasting models are explained in Section 4, followed by an explanation of the evaluation metrics in Section 5. Section 6 discusses the findings of the out-of-sample analysis and provides robustness checks. Finally, Section 7 concludes the paper.

2 Related Literature

Growth-at-Risk as a risk measure gained prominence following its adoption by institutions like the IMF (Prasad et al., 2019), making it a focal point for research on economic tail risks modeled through predictive quantile regressions (see, for example, De Nicolò and Lucchetta, 2017; Adams et al., 2021; Amburgey and McCracken, 2023). In particular,

due to the influential paper by Adrian et al. (2019), much of the empirical literature focuses on the negative correlation between financial conditions and economic downside risks. They find that lower quantiles of the conditional GDP growth distribution vary significantly with financial conditions, whereas the upper part of the distribution remains relatively stable over time. This time-varying asymmetry led to the conclusion that GDP growth exhibits negative skewness during recessions and is closer to symmetric or even positively skewed during expansions (Delle Monache et al., 2024). Similarly, Giglio et al. (2016) provide evidence that systemic risk measures have an asymmetric effect on the conditional distribution of macroeconomic outcomes. However, rather than interpreting this as evidence in favor of time-varying skewness, they argue that systemic risk measures are more informative about the lower tail than the central tendency or upper tail of macroeconomic indicators.

Beyond financial conditions, the shape of the predictive distribution of GDP has also been shown to depend on the choice of conditioning variables and the methods employed. For instance, Carriero et al. (2022) use Bayesian mixed-frequency models with a broad set of macroeconomic indicators and find only a slight degree of asymmetry in GDP growth. Brownlees and Souza (2021), using a GARCH model and estimating the innovation distribution nonparametrically, similarly find that GDP growth is not significantly skewed and displays moderately fat tails. Moreover, the results by Adrian et al. (2019), namely that only the lower tail of GDP growth is time-varying, could also be explained by a symmetric distribution with time-varying mean and variance (Carriero et al., 2024). This implies that modeling GaR requires tracking conditional mean and variance, and potentially kurtosis, rather than relying on time-varying skewness alone (Fagiolo et al., 2008; Figueres and Jarociński, 2020; Plagborg-Møller et al., 2020).

There is also a more critical perspective on the role of financial conditions in predicting growth vulnerabilities. Reichlin et al. (2020) find that financial conditions add little to predicting the lower tail of GDP growth once real economic indicators are considered. Similarly, Plagborg-Møller et al. (2020) show that quantile regressions using both financial and macroeconomic variables offer only limited insights into the conditional mean of GDP growth at short horizons, with high uncertainty around higher moments and longer forecast horizons. Since financial condition indices like the NFCI strongly co-move with real variables tied to GDP growth, this raises concerns about their marginal predictive value. Yet, their strong correlation with real activity suggests they may still capture relevant information for modeling GaR.

This paper builds directly on the work of Brownlees and Souza (2021), who apply a GARCH model to a panel of countries for GaR forecasting. They find that a purely backward-looking univariate GARCH model outperforms quantile regressions including

the NFCI in forecasting the lower tail of GDP growth and delivers more accurate density forecasts overall. Their results highlight the advantage of modeling GDP volatility through GARCH dynamics, which remain agnostic about the source of downside risks and avoid reliance on potentially unstable predictors (Brownlees and Souza, 2021).

One drawback of the univariate GARCH model is its reliance solely on lagged GDP data. To address this, the present paper incorporates SPF forecasts into the conditional mean of an AR-GARCH model, drawing on a literature that uses historical forecast errors to assess uncertainty around survey-based consensus projections. For example, Reifschneider and Tulip (2019) describe how the Federal Open Market Committee (FOMC) constructs confidence bands around point projections using root mean squared forecast error estimates from the Summary of Economic Projections. These bands, however, assume forecast uncertainty is constant or evolves slowly. Clark et al. (2020) find that forecast error variance exhibits time variation, and modeling it via stochastic volatility improves SPF-based density forecasts. Similarly, Adams et al. (2021) show that conditioning SPF projections on financial conditions (e.g., NFCI) improves density forecast accuracy by capturing variations in uncertainty.

This paper contributes to that strand by leveraging SPF median forecasts to nowcast GaR, building on evidence that consensus forecasts offer accurate short-term predictions. While individual survey forecasts can be biased (Zarnowitz, 1985; Elliott et al., 2008), consensus projections often outperform standard time-series models, especially at shorter horizons (Ang et al., 2007; Giannone et al., 2008; Bańbura et al., 2013; Bańbura et al., 2021). For instance, Stark (2010) show that SPF median forecasts outperform model-based estimates for current-quarter GDP, although accuracy deteriorates rapidly for longer horizons.

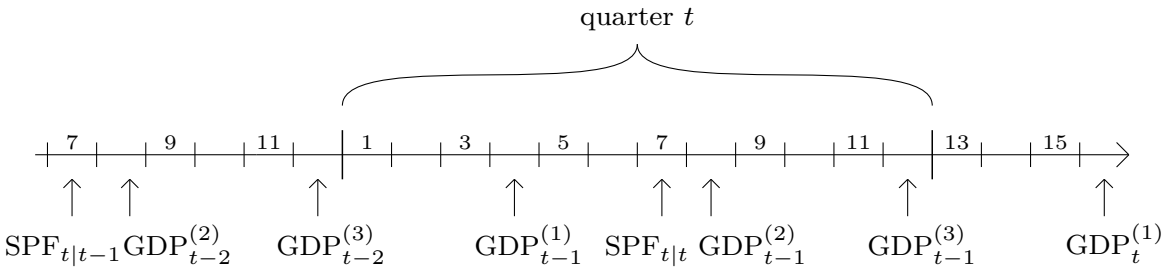
Although the SPF also provides density forecasts through histograms, which have been used to assess tail risks (Andrade et al., 2014; Ganics et al., 2024), this paper does not use them for several reasons. These forecasts refer to fixed events (e.g., annual GDP), and must be transformed into fixed-horizon forecasts. Ganics et al. (2024) address this by combining densities, but their method requires fitting continuous distributions to histogram bins, a process complicated by changing bin widths, unbounded tails, rounding, and sparse probability mass in some bins (Bassetti et al., 2023). Moreover, subjective density forecasts often suffer from misspecification and poor calibration (Rossi and Sekhposyan, 2019). Krüger and Plett (2024) find that forecast error-based approaches yield better interval forecasts than the SPF's density estimates. Also, mean forecasts derived from SPF histograms are typically less accurate and updated less frequently than point forecasts (Clements, 2010, 2014), and may reflect forecasters' confidence rather than true outcome distributions (Bańbura et al., 2021).

Finally, this paper relates to the broader literature on real-time nowcasting of GDP, which uses incoming data to update forecasts throughout the quarter (Giannone et al., 2008; Bańbura et al., 2013). High-frequency financial data has been shown to improve nowcasts of both central tendencies (Andreou et al., 2013) and tail risks (Ferrara et al., 2022). In the context of GaR, Carriero et al. (2022) show that using a broad set of macroeconomic indicators in real time enhances nowcast accuracy. For out-of-sample evaluation, real-time GDP vintages and SPF data are available (Clements et al., 2023). However, real-time NFCI data only became available in 2011. Much of the literature relies on the final vintage of the NFCI, which could bias conclusions about its predictive value (see, e.g., Adrian et al., 2019; Reichlin et al., 2020; Brownlees and Souza, 2021; Carriero et al., 2022)

3 Real-time data and predictors

This paper focuses on real-time nowcasting of Growth-at-Risk in quarter t, necessitating the use of real-time data to avoid look-ahead bias. Given the delayed publication of GDP data, Figure 1 provides an overview of the releases and revisions of quarterly GDP.

Figure 1: Release calendar



Notes: This graph sketches the release and revision dates of GDP together with the release dates of the SPF forecasts. The numbers on the time line indicate the weeks in each quarter.

In the fourth week of quarter t, the Bureau of Economic Analysis (BEA) releases the first estimate of the previous quarter's GDP, denoted by $\text{GDP}^{(1)}_{t-1}$. Consequently, during the early weeks of quarter t, the GDP of $t-1$ remains unavailable. Approximately one and two months after the initial release, the BEA issues the first and second major revisions, leading to the second ($\text{GDP}^{(2)}_{t-1}$) and third ($\text{GDP}^{(3)}_{t-1}$) estimate of GDP, respectively. Analogously, the initial estimate for current-quarter GDP, $\text{GDP}^{(1)}_{t}$, is published at the end of the first month of quarter $t+1$.

The release calendar depicted in Figure 1 also shows release dates of the SPF. The SPF is a quarterly survey of macroeconomic forecasts for the US. Starting in the fourth quarter of 1968, the survey was initially conducted by the American Statistical Association

and the National Bureau of Economic Research. Since 1990, the Federal Reserve Bank of Philadelphia has been running the survey that consists of roughly 40 professional forecasters (Clements et al., 2023). For GDP, based on the median SPF projections, the Philadelphia Fed provides forecasts for the annualized percentage points quarter-over-quarter growth rates. This includes nowcasts for the current quarter and up to forecasts of one-year ahead quarterly growth rates. Usually, the survey is released mid-quarter, i.e., around the sixth to seventh week of a quarter.

Table 1 summarizes the latest available information in real-time on a weekly basis, omitting even numbered weeks to save space. To reconstruct the information set available in each week of quarter t, I take monthly vintages of real GDP from the Real-Time Data Set for Macroeconomists of the Philadelphia Fed and match them with the historical release dates available on ALFRED.[1] For quarterly GDP, in week w of quarter t, let $y_{s,w}$ denote the annualized quarterly log growth rates for quarters $s < t$ according to

$$y_{s,w} = y_{s|t,w} = 400 \cdot (\log(\text{GDP}_{s|t,w}) - \log(\text{GDP}_{s-1|t,w})), \tag{1}$$

where $\text{GDP}_{s|t,w}$ denotes real GDP in quarter s obtained from the monthly vintage available in week w of quarter t. Due to delayed releases of GDP, typically in the first three weeks of quarter t, GDP for quarter $t-1$ is not available, so growth rates are only obtained for $s = 1, \ldots, t-2$. In the following weeks, a forecaster has information on GDP for $s = 1, \ldots, t-1$, with a first revision, $\text{GDP}^{(2)}_{t-1}$, occurring between the 7th and 9th week.

Concerning the SPF, at any given moment and across forecast horizons, the survey projections are real-time forecasts. To reconstruct the information flow provided by the SPF thus only requires knowledge about the timing of the survey. Since the Fed is in charge of the SPF, release dates are available on the web page of the Philadelphia Fed, permitting an exact alignment with the available information set.[2] Typically, in the first five to six weeks of each quarter, only the one-step-ahead projections, $\text{SPF}_{t|t-1}$, are available. Consequently, the nowcasts, $\text{SPF}_{t|t}$, can generally only be exploited in the second half of the quarter. To facilitate notation, let $SPF_{s,w}$ denote the SPF median projections for $s = 1, \ldots, t$, available in week $w = 1, \ldots, 15$ of quarter t according to

$$SPF_{t,w} = \begin{cases} \text{SPF}_{t|t-1} & \text{if } w \leq 6 \\ \text{SPF}_{t|t} & \text{if } w > 6. \end{cases} \tag{2}$$

[1] The BEA reported GNP until the end of 1991. Thus, for a substantial part of the sample, GDP refers to GNP.

[2] For the first quarter in 1990, the Federal Reserve Bank of Philadelphia is uncertain about the release date. The release date is thus chosen to be in the seventh week of the first quarter of 1990. However, this only affects the first quarter of the out-of-sample evaluation.

Table 1: Real-time information available for predicting GDP growth in quarter t

week (w)	Variables
1	$\text{GDP}^{(3)}_{t-2} = \text{GDP}_{t-2\|t,12}$, $\text{SPF}_{t\|t-1}$, $\text{NFCI}_{t-1,12\|t,1}$
3	$\text{GDP}^{(3)}_{t-2} = \text{GDP}_{t-2\|t,12}$, $\text{SPF}_{t\|t-1}$, $\text{NFCI}_{t,2\|t,3}$
5	$\text{GDP}^{(1)}_{t-1} = \text{GDP}_{t-1\|t,4}$, $\text{SPF}_{t\|t-1}$, $\text{NFCI}_{t,4\|t,5}$
7	$\text{GDP}^{(1)}_{t-1} = \text{GDP}_{t-1\|t,4}$, $\text{SPF}_{t\|t}$, $\text{NFCI}_{t,6\|t,7}$
9	$\text{GDP}^{(2)}_{t-1} = \text{GDP}_{t-1\|t,8}$, $\text{SPF}_{t\|t}$, $\text{NFCI}_{t,8\|t,9}$
11	$\text{GDP}^{(2)}_{t-1} = \text{GDP}_{t-1\|t,8}$, $\text{SPF}_{t\|t}$, $\text{NFCI}_{t,10\|t,11}$
13	$\text{GDP}^{(3)}_{t-1} = \text{GDP}_{t-1\|t,12}$, $\text{SPF}_{t\|t}$, $\text{NFCI}_{t,12\|t+1,1}$
15	$\text{GDP}^{(3)}_{t-1} = \text{GDP}_{t-1\|t,12}$, $\text{SPF}_{t\|t}$, $\text{NFCI}_{t+1,2\|t+1,3}$

Notes: This table demonstrates the latest realizations of GDP, SPF, and the NFCI available for odd numbered weeks throughout quarter t and the first three weeks of quarter $t+1$.

Note that the timing of the SPF release may vary, so the transition from one-step-ahead projections to nowcasts might occur earlier or later than week 6.

Lastly, Table 1 displays data releases of the Chicago Fed's NFCI. The NFCI is a weekly estimate of financial conditions in the US, incorporating a comprehensive set of financial indicators including money markets, debt and equity markets, and the banking system. Its construction is described in detail in Brave and Butters (2012). The index is standardized to have a mean of zero and a standard deviation of one, with positive readings historically associated with tighter than average financial conditions, and vice versa.

Official real-time vintages of the NFCI are obtainable on ALFRED. As the first vintage became available in 2011, many studies employ the final vintage of the NFCI. However, pseudo out-of-sample analyses ignore the filtering uncertainty that primarily affects the real-time NFCI at sample endpoints (Brownlees and Souza, 2021). Moreover, the NFCI incorporates factors directly dependent on GDP, such as the corporate debt-to-GDP ratio, which can introduce look-ahead bias when used in pseudo out-of-sample analyses. Therefore, I rely on unofficial real-time weekly vintages of the NFCI constructed by Amburgey and McCracken (2023). These unofficial vintages begin in 1988 and include NFCI data dating back to 1973.

The NFCI is typically released on Wednesdays, incorporating data up to the preceding Friday, resulting in a one-week publication delay. Considering the quarterly frequency of GDP data, a simple aggregation scheme is to align weekly NFCI data in week w at a

quarterly frequency such that

$$NFCI_{s,w} = \text{NFCI}_{s,w-1|t,w}, \tag{3}$$

where $\text{NFCI}_{s,w-1|t,w}$ is the NFCI in week $w-1$ of quarter s, available from the latest NFCI vintage in week w of quarter t. For instance, in the first week of quarter t, $NFCI_{s,1} = \text{NFCI}_{s-1,12|t,1}$, whereas in the second week, $NFCI_{s,2} = \text{NFCI}_{s,1|t,2}$, and so on.

4 Forecasting models

The aim is to nowcast Growth-at-Risk for quarter t in real-time. Therefore, GaR is re-estimated on a weekly frequency, taking into account the latest information available at the end of each week. The nowcasting exercise starts in the first fully observed week of the quarter t and ends in the third week of the subsequent quarter $t+1$. Thus, for each quarter t through the evaluation period, there are 15 weeks considered. The release of the first estimate of GDP is usually in the fourth week of quarter $t+1$. Therefore, the real-time data and the associated nowcasts are updated up to the latest releases and revisions just before the Bureau of Economic Analysis publishes the first estimate of GDP for quarter t. This approach to timing is identical to the one of Carriero et al. (2022), ensuring that the forecast horizons are comparable across quarters.

4.1 GARCH-type models

For GaR prediction, Brownlees and Souza (2021) find that the GARCH(1,1) model and quantile regressions exhibit similar out-of-sample performance. Therefore, I consider AR-GARCH-type models to nowcast quantiles of the GDP growth distribution. The conditional mean and variance of $y_{t,w}$ are modeled as

$$\mu_{t|t,w} = \phi_{0,w} + \phi_{1,w} y_{t-j,w} + x^{\mu}_{t,w}\delta_w \tag{4}$$

$$\sigma^2_{t|t,w} = \omega_w + \alpha_w \varepsilon^2_{t-j,w} + \beta_w \sigma^2_{t-j|t,w} + \exp(x^{\sigma}_{t,w}\gamma_w), \tag{5}$$

where $\varepsilon_{s,w} = y_{s,w} - \mu_{s|t,w}$, $\omega_w > 0$, $\alpha_w > 0$, $\beta_w > 0$, $\alpha_w + \beta_w < 1$, and $j \in \{1, 2\}$, depending on the release date of previous quarters' GDP. Thus, all GARCH-type specifications considered are of the form AR(1)–$x^{\mu}_{t,w}$–GARCH(1,1)–$x^{\sigma}_{t,w}$, where $x^{\mu}_{t,w}$ and $x^{\sigma}_{t,w}$ represent vectors of additional explanatory variables for the conditional mean and the conditional variance, respectively. The elements in $x^{\sigma}_{t,w}$ are standardized to have mean zero and variance one.

Assuming normality, the conditional distribution of GDP growth in week w is

$$y_{t|t,w} \sim \mathcal{N}(\mu_{t|t,w}, \sigma^2_{t|t,w}). \tag{6}$$

Equation (4) resembles a linear regression of $y_{t,w}$ on a constant, one lag of GDP growth, and additional predictors in $x^{\mu}_{t,w}$. Similarly, equation (5) corresponds to a GARCH(1,1) specification, allowing for potential predictors in $x^{\sigma}_{t,w}$. To ensure a strictly positive conditional variance, explanatory variables for $\sigma^2_{t|t,w}$ are incorporated through the exponential function, $\exp(\cdot)$.

In the baseline specification (henceforth AR-GARCH), both $x^{\mu}_{t,w}$ and $x^{\sigma}_{t,w}$ are omitted, resulting in a purely univariate time series model with an AR(1) specification for the conditional mean. When incorporating real-time median SPF projections in equation (4), i.e., $x^{\mu}_{t,w} = SPF_{t,w}$, the model is denoted as SPF-GARCH.[3] Regarding the NFCI, it is uncertain whether including it in the conditional mean or variance would refine GaR nowcasts. Thus, the out-of-sample predictive efficacy of the NFCI is evaluated as a regressor in both the conditional mean and the conditional variance.

The GARCH model parameters in equations (4) and (5) are estimated by quasi-maximum likelihood. Mean predictions, $\hat{\mu}_{t|t,w}$, are readily available from estimates on the parameters in equation (4). Conditional quantiles and density forecasts can be obtained from estimates on the conditional volatility, $\hat{\sigma}^2_{t|t,w}$, in equation (5) and assuming normality in equation (6). In week w of quarter t, the nowcast of the conditional τ-quantile, where $\tau \in (0, 1)$, is then given by

$$\hat{Q}^{\text{GARCH}}_{t|t,w}(\tau) = \hat{\mu}_{t|t,w} + \sqrt{\hat{\sigma}^2_{t|t,w}} F^{-1}(\tau), \tag{7}$$

where $F^{-1}(\cdot)$ is the inverse cumulative distribution function of $\mathcal{N}(0, 1)$. Due to the small sample sizes in the out-of-sample analysis, assuming a standard normal distribution for the innovations is intended to achieve a parsimonious parameterization. This assumption can be relaxed, for instance, by modeling a more flexible distribution that incorporates skewness and fat tails or by estimating the distribution semi-parametrically through filtered historical simulation.

4.2 Quantile regressions

As popularized by Adrian et al. (2019), among others, in the GaR literature, quantile regressions of Koenker and Bassett (1978) are a commonly used method to predict

[3] Choosing to include the median instead of the mean SPF projections is arbitrary but more robust against extreme individual forecasts. However, in the robustness section 6.5.1, employing the mean SPF forecasts yields nearly identical out-of-sample results for GaR and density forecasts.

conditional quantiles of the GDP growth distribution. At time T, for the conditional τ-quantile, the vector of quantile regression parameters β_w^τ is estimated by minimizing the asymmetric absolute loss function

$$\hat{\beta}_w^\tau = \underset{\beta_w^\tau \in \mathbb{R}^k}{\arg\min} \sum_{t=1}^{T-j} (y_{t,w} - x_{t,w}\beta_w^\tau)(\tau - \mathbb{1}_{\{y_{t,w} - x_{t,w}\beta_w^\tau < 0\}}), \tag{8}$$

where $y_{t,w}$ represents GDP growth rates available in week w of quarter T, $x_{t,w}$ is a vector of predictors, $\mathbb{1}_{(\cdot)}$ is the indicator function, and j depends on the release date of previous quarters' GDP, taking values in $\{1, 2\}$. Thus, in weeks $w = 1, \ldots, 3$, the sum runs from $t = 1$ to $T - 2$, and in weeks $w = 4, \ldots, 15$, it runs from $t = 1$ to $T - 1$. The index w emphasizes that the quantile regressions are re-estimated each week. Then, the predicted τ-quantile of GDP growth in quarter t conditional on data up to week w reads

$$\hat{Q}^{\text{QR}}_{t|t,w}(\tau) \equiv \hat{Q}^{\text{QR}}_{y_{t,w}}(\tau|x_{t,w}) = x_{t,w}\hat{\beta}_w^\tau. \tag{9}$$

Thus, for each quarter t through the evaluation period, there are 15 nowcasts $\hat{Q}^\tau_{t|t,w}$, for $w = 1, \ldots, 15$.

In the baseline specification, the autoregressive quantile regression (QR-AR) only includes a constant and one lag of $y_{t,w}$, i.e., $x_{t,w} = (1, y_{t-j,w})$. If $x_{t,w}$ also includes the $NFCI_{t,w}$ ($SPF_{t,w}$), this specification is referred to as QR-NFCI (QR-SPF).

5 Forecast evaluation

I define the "true" outcomes of GDP growth following the approach of Carriero et al. (2022) and others. This involves using GDP growth rates available one quarter after the initial release of GDP. Thus, actual GDP growth is defined as annualized quarterly log growth rates according to

$$y_t = 400 \cdot (\log(\text{GDP}_{t|t+2,1}) - \log(\text{GDP}_{t-1|t+2,1})), \tag{10}$$

where $\text{GDP}_{t|t+2,1}$ denotes the latest revised version of GDP for quarter t available in the first week of quarter $t + 2$, typically corresponding to the third monthly release, $\text{GDP}_t^{(3)}$. Therefore, identical to Carriero et al. (2022), the actual GDP growth rates are given by the second releases in the quarterly real-time vintages available at the Fed database.

For evaluating conditional quantile nowcasts, the quantile score (QS) is a strictly consistent scoring rule (Gneiting, 2011). The asymmetric piecewise linear loss function for

the τ-quantile in week w of quarter t is computed as

$$\text{QS}^{\tau}_{t,w} = (\mathbb{1}_{\{\hat{Q}_{t|t,w}(\tau) \geq y_t\}} - \tau)(\hat{Q}_{t|t,w}(\tau) - y_t), \tag{11}$$

where y_t denotes actual GDP. To evaluate predictive distributional forecasts, the Continuous Ranked Probability Score (CRPS) is strictly consistent and is approximated by

$$\text{CRPS}_{t,w} = \frac{2}{J-1} \sum_{j=1}^{J-1} \hat{Q}_{t|t,w}(\tau_j), \tag{12}$$

where $\tau_j = j/J$ and $J = 20$ (Gneiting and Raftery, 2007).

Statistical inference on the out-of-sample predictive performance requires comparing many models against each other. Instead of testing each competitor model against a benchmark model, say QR-AR, I report the model confidence set (MCS) of Hansen et al. (2011) based on the Diebold and Mariano (1995) test to account for the problem of multiple comparisons. This approach has the advantage of jointly testing the null hypothesis of equal predictive performance of the considered models. Let $\mathcal{M}_0$ denote the set of all competing models indexed by $k = 1, \ldots, K$. Then, the relative performance of two models, k and l, is measured by the loss differential, $d^w_{kl,t} \equiv L^k_{t,w} - L^l_{t,w}$, at week w through quarters t, for all $k, l \in \mathcal{M}_0$. Here, the loss functions considered are either $\text{QS}^{\tau}_{t,w}$ or $\text{CRPS}_{t,w}$. The null hypothesis is $H_0 : \mathbb{E}(d^w_{kl,t}) = 0$ for all $k, l \in \mathcal{M}$, where $\mathcal{M} \subset \mathcal{M}_0$. The $1 - \alpha\%$ MCS is defined as $\mathcal{M}^{\star} \equiv \{k \in \mathcal{M}_0 : \mathbb{E}(d^w_{kl,t}) \leq 0 \text{ for all } l \in \mathcal{M}_0\}$. The test statistic, $T_{\mathcal{M}} = \max_{k,l \in \mathcal{M}} |t^w_{kl}|$, is based on the average loss difference

$$t^w_{kl} = \frac{\overline{d}^w_{kl}}{se(\overline{d}^w_{kl})} \text{ for all } k, l \in \mathcal{M}, \tag{13}$$

where $\overline{d}^w_{kl}$ and $se(\overline{d}^w_{kl})$ are the mean and standard error of the loss differentials $d^w_{kl,t}$ over quarters t. Thus, the test sequentially eliminates the worst-performing model from $\mathcal{M}_0$ if $T_{\mathcal{M}}$ exceeds the critical value, indicating a rejection of equal performance. Since the asymptotic distribution of $T_{\mathcal{M}}$ is nonstandard, it is simulated by block-bootstrapping, where I use 15.000 replications. As barely any autocorrelations of loss differentials are significant after the second lag, the block length chosen is 3. However, the results are not sensitive to these settings.[4]

One critical aspect is the limiting distribution of $T_{\mathcal{M}}$ if models in $\mathcal{M}_0$ are nested. Specifically, for this distribution to exist, $d^w_{kl,t}$ must be stationary with strictly positive variance. Thus, a common approach is to estimate the models on a rolling window,

[4]The MCS procedure is implemented in the MFE Toolbox of Sheppard (2009), available at `https://www.kevinsheppard.com/code/matlab/mfe-toolbox/`.

ensuring a well-behaved asymptotic distribution with the variance of $d^w_{kl,t}$ bounded away from zero (Giacomini and White, 2006). However, the use of quarterly data results in considerable estimation uncertainty due to small sample sizes. Consequently, I employ expanding window estimation to reduce estimation uncertainty and thus increase power of the MCS. Additionally, robustness checks using rolling window estimation show that the results remain qualitatively unchanged.

In determining the evaluation period, a trade-off exists between selecting a later starting point for higher estimation accuracy and choosing an earlier starting point to increase the test procedure's power. Given the SPF's inception in 1968, the evaluation period spans from the first quarter of 1990 to the fourth quarter of 2019, resulting in 85 in-sample observations at the beginning of the evaluation period. Excluding the onset of the COVID-19 pandemic yields 120 out-of-sample observations to assess the models' prediction accuracy. Additionally, to evaluate the predictive performance during the pandemic, the analysis extends to the subsequent two years. For the evaluation, I report the 90% MCS, conducting tests of equal predictive performance at the 10% significance level.

6 Empirical results

This section presents the empirical results on the accuracy of the real-time nowcasts of Growth-at-Risk. In the following, I exclude the COVID-19 pandemic to ensure that one extreme outlier is not driving the results. Later, I will extend the sample and moreover study two periods of economic turmoil separately. Excluding the pandemic, the evaluation starts in the first quarter of 1990 and ends in the fourth quarter of 2019, resulting in 120 out-of-sample observations.

Before turning to quantile- and density forecasts, I provide motivating evidence that supports exploiting the information content provided by the SPF median forecasters and updating data in real-time. Regarding the SPF forecasting performance, Table 2 shows summary statistics of the forecast errors of the SPF median projections. The average one-step-ahead forecast error of the SPF is close zero, whereas the nowcasts seem to be too optimistic on average. However, for both forecast horizons, the forecast errors are not significantly different from zero, suggesting that unbiasedness of the SPF median forecasts cannot be rejected. In addition, the root mean squared forecast error (RMSFE) of the nowcasts is significantly smaller than of the one-step-ahead forecasts (p-value = 0.06). Thus, the forecast accuracy of the SPF increases for shorter horizons.

The lower panel reports estimates for the Mincer and Zarnowitz (1969) (MZ) regression, i.e., a regression of actual GDP growth on the SPF projections, resulting in coefficient estimates on the constant, ρ_0, and the SPF median forecasts, ρ_1. Jointly testing $\rho_0 = 0$

Table 2: SPF unbiasedness and efficiency

SPF error	One-step-ahead		Nowcasts	
mean	-0.035	(0.200)	0.220	(0.144)
RMSFE	1.850		1.557	
MZ regression	**One-step-ahead**		**Nowcasts**	
ρ_0	-0.658	(0.816)	0.027	(0.340)
ρ_1	1.252	(0.292)	1.087	(0.132)
R^2	0.280		0.492	
p-value	0.680		0.220	

Notes: The upper panel reports the mean forecast error and the root mean squared forecast error (RMSFE) of the Survey of Professional Forecasters (SPF). The lower panel presents estimates from Mincer–Zarnowitz (MZ) regressions, with Newey–West standard errors in parentheses. Reported p-values correspond to the joint null hypothesis $H_0 : \rho_0 = 0,\ \rho_1 = 1$. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

and $\rho_1 = 1$ reveals that unbiasedness and efficiency of the SPF median forecasts cannot be rejected. In particular, for the nowcasts, the constant is close to zero and the slope coefficient is close to unity. Moreover, the R^2 increases from 0.28 to 0.49. Thus, the SPF consensus point predictions have desirable properties, especially for short horizons.

Next, to motivate utilizing the real-time information flow as GDP and SPF data are released, Table 3 presents the RMSFE for different forecast origins within a quarter, starting from the first fully observed week. Specifically, I consider nowcasts of the conditional mean of GDP growth based on different AR-GARCH type specifications, with the univariate AR-GARCH model solely relying on past releases of GDP growth. Gray areas in the table indicate the 90% MCS.

Table 3: Out-of-sample evaluation of mean forecasts

	week							
RMSFE	1	3	5	7	9	11	13	15
AR-GARCH	2.21	2.19	2.02	2.02	1.99	1.99	2.01	2.01
$+x^{\mu}_{t,w} = SPF_{t,w}$	**1.96**	**1.95**	**1.89**	**1.65**	**1.60**	**1.60**	**1.60**	**1.60**
$+x^{\mu}_{t,w} = NFCI_{t,w}$	2.18	2.20	1.95	1.99	1.98	1.99	1.95	2.01

Notes: This table reports the root mean squared forecast errors (RMSFE) over the out-of-sample evaluation period. Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set based on the mean squared error loss, and bold values highlight the lowest RMSFE within each column. The AR-GARCH model refers to the AR(1)–GARCH(1,1) specification, while $NFCI_{t,w}$ denotes the Chicago Fed National Financial Conditions Index. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

First, the RMSFE of the purely autoregressive AR-GARCH model tends to become smaller with decreasing forecast horizon, demonstrating that incorporating the latest releases and revisions of GDP data leads to improvements in nowcasting GDP growth. The reduction in the average forecast errors is most prominent between the third and fifth weeks, associated with the publication of the first estimate of the previous quarter's GDP. Later revisions published between weeks 7 and 9 and between weeks 11 and 13 appear less informative.

Second, adding the SPF to the conditional mean of the AR-GARCH mostly significantly reduces out-of-sample forecast errors. In the first four weeks, the SPF-GARCH model exploits the one-step-ahead median SPF forecast. Apart from a few exceptions, the release of the SPF nowcast occurs between the fifth and seventh weeks. The results reveal that the publication of the SPF median nowcast strongly reduces the RMSFE. Thus, accounting for the latest releases of both GDP and SPF reduces forecast errors. Interestingly, the AR-GARCH model, including the SPF as an additional regressor, results in larger losses than the SPF median projections shown in Table 2. The increased estimation uncertainty of the GARCH model translates into slightly less accurate forecasts of GDP growth, however, unreported results show that these differences are not significant.

Lastly, adding the NFCI into the conditional mean does not improve forecast accuracy, indicating that financial conditions are not informative about the center of the conditional GDP growth distribution.

6.1 SPF versus NFCI

To predict Growth-at-Risk as well as the conditional density of GDP growth, the models considered are GARCH and quantile regressions. In order to evaluate downside and upside risk, I report the average 10% and 90% quantile scores. The predictive density is assessed by the CRPS. Table 4 reports average losses of the quantile scores and the CRPS, along with the 90% MCS for different forecast origins within a quarter, starting in the first fully observed week. The upper panel shows results for 10% quantile nowcasts of GDP growth.

Both univariate specifications, AR-GARCH and QR-AR, exhibit increasing nowcast accuracy as GDP releases and revisions become available during the quarter. Notably, there are two distinct declines in average losses observed between weeks 3 to 5 and between weeks 7 to 9. The former coincides with the release of the first estimate of GDP for the previous quarter, while the latter is attributable to its first revision. Moreover, the AR-GARCH model performs better than the QR-AR approach.

Incorporating the NFCI into quantile regressions underscores its substantial predictive power for downside risk. The results demonstrate that quantile regressions utilizing the NFCI as an explanatory variable perform best in the first weeks. While this finding is

Table 4: Out-of-sample evaluation using SPF or NFCI

	week							
10% quantile score	1	3	5	7	9	11	13	15
AR-GARCH	0.44	0.43	0.40	0.40	0.37	0.37	0.37	0.37
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.40	0.38	0.36	0.36	0.35	0.35	0.34	0.34
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.41	0.39	0.37	0.36	0.34	0.34	0.34	0.33
$+x^{\mu}_{t,w} = SPF_{t,w}$	0.36	0.35	0.35	**0.28**	**0.29**	**0.29**	**0.29**	**0.29**
QR-AR	0.50	0.51	0.49	0.49	0.47	0.47	0.47	0.47
$+NFCI_{t,w}$	**0.31**	**0.32**	**0.32**	0.32	0.30	0.33	0.32	0.33
$+SPF_{t,w}$	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
90% quantile score	1	3	5	7	9	11	13	15
AR-GARCH	0.40	0.40	0.35	0.35	0.36	0.36	0.36	0.36
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.41	0.42	0.38	0.38	0.37	0.39	0.37	0.37
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.42	0.41	0.38	0.38	0.39	0.39	0.39	0.38
$+x^{\mu}_{t,w} = SPF_{t,w}$	**0.34**	**0.34**	**0.34**	**0.31**	**0.31**	**0.31**	**0.31**	**0.31**
QR-AR	0.55	0.54	0.46	0.46	0.46	0.46	0.47	0.46
$+NFCI_{t,w}$	0.46	0.45	0.39	0.37	0.37	0.38	0.39	0.40
$+SPF_{t,w}$	0.39	0.39	0.38	0.36	0.33	0.33	0.33	0.33
CRPS	1	3	5	7	9	11	13	15
AR-GARCH	1.25	1.24	1.14	1.14	1.13	1.13	1.14	1.14
$+x^{\mu}_{t,w} = NFCI_{t,w}$	1.27	1.28	1.15	1.17	1.16	1.17	1.15	1.17
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	1.28	1.25	1.17	1.17	1.16	1.15	1.16	1.15
$+x^{\mu}_{t,w} = SPF_{t,w}$	**1.12**	**1.12**	**1.09**	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
QR-AR	1.35	1.34	1.25	1.25	1.24	1.24	1.25	1.25
$+NFCI_{t,w}$	1.22	1.24	1.17	1.17	1.18	1.18	1.18	1.19
$+SPF_{t,w}$	1.17	1.17	1.14	1.03	1.00	1.00	1.00	1.00

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average
specification, while SPF-GARCH augments the conditional mean with consensus forecasts from the SPF. The QR-AR model refers to a quantile regression with an AR(1) specification. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

not new, it is obtained without look-ahead bias and confirms previous research (see, for example, Giglio et al., 2016; Adrian et al., 2019; Ferrara et al., 2022).

In quantile regressions, the NFCI serves directly as a predictor for the targeted quantile. In the GARCH model, assuming normality, the NFCI predicts downside risk by explaining changes in the conditional mean or variance. Incorporating the NFCI into either the conditional mean, $x^{\mu}_{t,w}$, or the conditional variance, $x^{\sigma}_{t,w}$, reduces the nowcast errors compared to the basic AR-GARCH model, contrasting the findings by Brownlees and

Souza (2021). However, at the beginning of the quarter, the AR-GARCH with NFCI in the conditional mean or variance yields higher quantile losses than the QR-NFCI model. Towards the end of the quarter, the GARCH models incorporating NFCI, whether in the conditional mean or volatility, demonstrate a predictive performance similar to that of the QR-NFCI model.

The most striking result of Table 4 is the inclusion of the SPF-GARCH model in the MCS. In the first five weeks, the SPF-GARCH model utilizes the one-step-ahead median SPF forecasts in the conditional mean. Throughout this period, the QR-NFCI exhibits lower quantile losses. As the SPF nowcasts become available between weeks 5 and 7, in the second half of the quarter, the SPF-GARCH significantly outperforms its competitors, except for the QR-NFCI approach in weeks 7 to 9. The SPF median forecasts also improve GaR predictions in quantile regressions. However, the MCS does not include the QR-SPF model, although in the last weeks of the nowcasting exercise, the QR-SPF model exhibits similar average quantile losses. This suggests that quantile regressions do not exploit the information flow stemming from the SPF median forecasts for predicting downside risk as effectively as the GARCH model does.

Examining the 90% quantile losses, the middle panel of Table 4 assesses the predictive performance concerning upside risk. The SPF notably enhances the 90% quantile nowcasts for both the GARCH model and quantile regressions. However, employing median SPF forecasts in the GARCH model is statistically superior compared to quantile regressions in forecasting the upper quantiles of the GDP growth distribution. While the NFCI also decreases the out-of-sample losses in quantile regressions, its effect is less pronounced compared to the SPF. Interestingly, the QR-NFCI model performs worse than the purely backward-looking AR-GARCH model. Moreover, when explaining changes in the mean and variance in the GARCH model, the NFCI appears uninformative, contributing mainly to estimation uncertainty in the baseline AR-GARCH model. This finding suggests that although the NFCI serves as a strong predictor for downside risk, it is not capable of explaining variation in the upper part of the GDP growth distribution, in line with Giglio et al. (2016).

Finally, the bottom panel of Table 4 examines the CRPS, evaluating the predictive performance across the entire distribution. Given the SPF's enhancement of both lower and upper quantile predictions in the GARCH model, it is unsurprising that the SPF-GARCH model significantly outperforms other models in predicting the conditional density of GDP growth. Moreover, the SPF median forecasts improve the predictive performance of quantile regressions. It is important to note that the median SPF projections focus on the center of the distribution. Consequently, using the SPF nowcasts directly as an explanatory variable for the conditional mean calibrates the entire distributional forecast through mean

changes. Conversely, in quantile regressions, the weight of the SPF median nowcast is determined by different quantile regression coefficients for each quantile separately. As the SPF-GARCH model yields more precise predictive densities, the former approach proves to be more effective in this context. Moreover, the AR-GARCH results in a sharper predictive density than the QR-NFCI approach, underscoring that the NFCI's predictive ability holds exclusively for lower quantiles of GDP growth.

This is also supported by the probability integral transform (PIT) plots in Figure A.1, which include confidence bands following Rossi and Sekhposyan (2019). The SPF one-step-ahead forecast (fourth week) and the nowcast (eight week) both result in well-calibrated predictive densities. In contrast, the QR-NFCI approach produces well-calibrated forecasts only for downside risks.

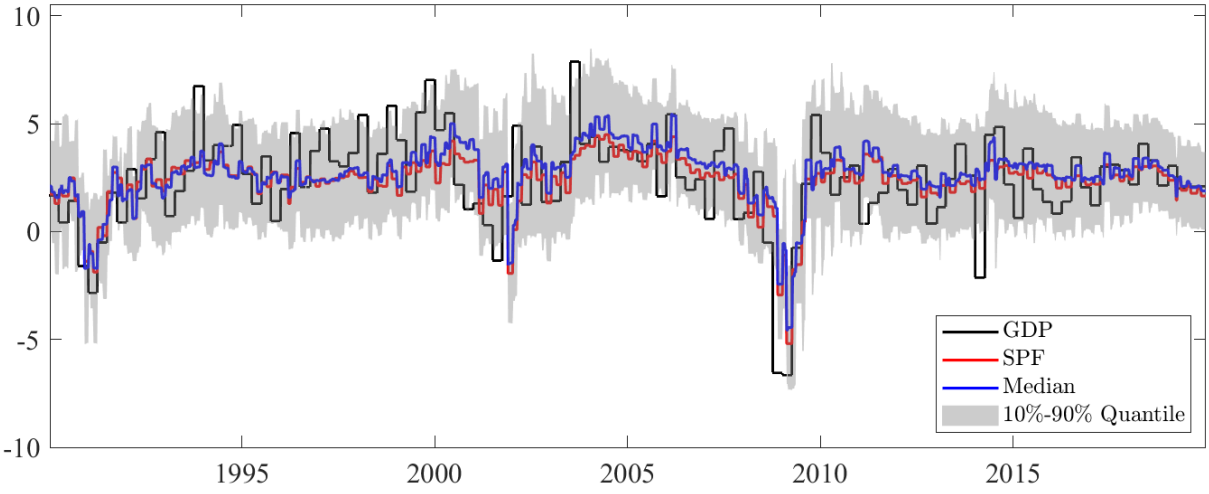
In summary, the findings highlight that the median SPF nowcasts are informative across all quantiles of GDP growth, with the SPF-GARCH model producing the most precise predictive densities. In the SPF-GARCH model, the conditional mean utilizes the SPF consensus nowcasts to capture variation in the central tendency of GDP growth, while the conditional volatility of the GARCH model helps generate a conditional density around the predicted mean of GDP growth. However, GARCH effects are strong, as indicated by β in the conditional variance in equation (5), which typically ranges between 0.8 and 0.9 throughout the out-of-sample period. Hence, incorporating the SPF median projections enhances predicting the distribution of GDP growth by providing accurate forecasts of the conditional mean, thus reducing past forecast errors, which alters the dynamics in the conditional volatility of the GARCH model. The GARCH model imposes more structure than quantile regressions, while the SPF captures structural changes, which turns out advantageous, particularly with small sample sizes. Additionally, while not in the MCS, quantile regressions incorporating the SPF provide the second-best density nowcasts, demonstrating superior predictive performance in comparison to the QR-NFCI approach.

Figure 2 illustrates the real-time predictive distribution of the SPF-GARCH model in panel (a) and the QR-NFCI approach in panel (b). Both models produce conditional 10% quantile nowcasts that closely track the patterns of quarterly GDP growth (black line). However, the QR-NFCI median nowcasts (blue line) appear to miss economic up- and downswings. In contrast, the SPF-GARCH model shows pronounced dynamics in the conditional mean of its predictive distribution, closely aligned with the SPF consensus forecasts (red line). In addition, the upper quantiles of the QR-NFCI are relatively constant over time, except for the aftermath of the global financial crisis around 2009, in line with Adrian et al. (2019). Notably, most of the variation observed in panel (b) originates from GDP releases when updating the nowcasts within a quarter. The time-variation

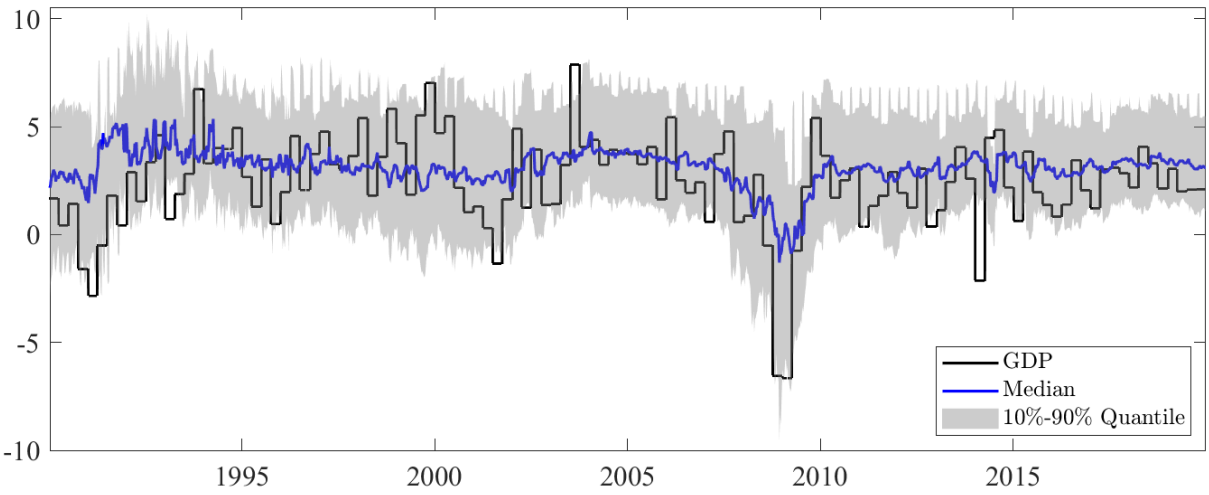
remains modest for a given week across quarters. In contrast, the predicted 90% quantiles of the SPF-GARCH model are less volatile within a quarter yet exhibit more variation across quarters, closely mirroring the GDP growth evolution. Lastly, as indicated by the significantly smaller CRPS, the density forecasts of the SPF-GARCH model are more closely centered around actual GDP growth than those of the QR-NFCI.

Figure 2: Real-time nowcasts

(a) SPF-GARCH



(b) QR-NFCI



Notes: This figure shows real-time nowcasts of quarterly US GDP growth over the out-of-sample evaluation period 1990:Q1 to 2019:Q4 for the SPF-GARCH model in panel (a) and the QR-NFCI approach in panel (b). The black line represents actual GDP growth, while the blue line shows the predicted median forecasts. The grey areas indicate density forecasts spanning the 10% to 90% conditional quantiles. In panel (a), the SPF-GARCH model assumes a normal distribution, so the median coincides with the mean, and the red line represents the SPF consensus forecasts.

6.2 SPF and NFCI combined

In the previous section, the SPF and the NFCI are considered separately as predictors, demonstrating that the SPF is superior over the entire distribution. The NFCI is informative about lower quantiles but less for the other parts of the distribution. However, in the first weeks of a quarter, the QR-NFCI model achieves the lowest scores at the 10% quantile of GDP growth. Thus, in this section, to assess whether the NFCI adds additional predictive gains to the SPF, both indicators are considered jointly.

To enhance comparability, I include the SPF-GARCH and the QR-SPF model from Section 6.1 and add the NFCI to these specifications to test for additional predictive gains. In addition, since the quantile regression with NFCI produces the lowest losses at the 10% quantile in the first weeks of a quarter, the QR-NFCI model is also included in this analysis.

Table 5 presents the out-of-sample results of this comparison. In the upper panel, for the 10% quantile of GDP growth, the QR-SPF model incorporating the NFCI performs best in the first weeks, while the SPF-GARCH model with NFCI in the conditional mean exhibits the lowest scores towards the end of the quarter. However, all models except the QR-SPF approach are contained in the MCS, indicating that the SPF-GARCH model is not significantly outperformed.

In terms of the 90% quantile, the SPF-GARCH model consistently provides the lowest out-of-sample losses, significantly outperforming the QR-NFCI approach. Moreover, adding the NFCI to the SPF-GARCH model does not result in predictive gains. Similarly, indicated by the CRPS, the SPF-GARCH model produces significantly more precise density forecasts than the QR-NFCI approach. Overall, across the entire distribution, the SPF-GARCH model is among the best performing models, with the addition of the NFCI showing no significant improvement in predictive accuracy.

6.3 Growth-at-Risk during economic turmoils

To test the robustness of the main findings, I extend the evaluation period through 2024:Q4. Given the extreme and unprecedented nature of the COVID-19 pandemic, univariate models perform poorly, reducing the power of the MCS procedure. Consequently, I exclude purely backward-looking models and focus on the SPF-GARCH model, along with the QR-SPF and QR-NFCI approaches. To better understand predictive performance during economic crises, I also examine the global financial crisis (GFC, 2007–2008) and the COVID-19 pandemic (2020–2022) in isolation.

Table 5: Out-of-sample evaluation using NFCI and SPF combined

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	0.28	0.29	0.29	0.29	0.29
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.37	0.35	0.34	0.28	0.27	**0.27**	**0.27**	**0.27**
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.35	0.34	0.33	0.28	0.28	0.28	0.28	0.28
QR-SPF	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
$+NFCI_{t,w}$	**0.29**	**0.30**	**0.29**	**0.28**	**0.27**	0.27	0.28	0.28
QR-NFCI	0.31	0.32	0.32	0.32	0.30	0.33	0.32	0.33
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	**0.34**	**0.34**	**0.34**	**0.31**	**0.31**	**0.31**	**0.31**	**0.31**
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.37	0.36	0.34	0.32	0.31	0.31	0.31	0.31
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.37	0.36	0.36	0.34	0.32	0.32	0.32	0.31
QR-SPF	0.39	0.39	0.38	0.36	0.33	0.33	0.33	0.33
$+NFCI_{t,w}$	0.41	0.39	0.36	0.35	0.34	0.34	0.34	0.34
QR-NFCI	0.46	0.45	0.39	0.37	0.37	0.38	0.39	0.40
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	**1.12**	**1.12**	**1.09**	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
$+x^{\mu}_{t,w} = NFCI_{t,w}$	1.18	1.16	1.09	1.01	0.97	0.97	0.97	0.98
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	1.14	1.14	1.10	1.01	0.98	0.98	0.98	0.98
QR-SPF	1.17	1.17	1.14	1.03	1.00	1.00	1.00	1.00
$+NFCI_{t,w}$	1.17	1.17	1.12	1.02	0.98	0.99	0.98	0.99
QR-NFCI	1.22	1.24	1.17	1.17	1.18	1.18	1.18	1.19

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with consensus forecasts from the SPF in the conditional mean. The QR-SPF model refers to a quantile regression augmented with SPF consensus forecasts. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

6.3.1 Including the COVID-19 pandemic

Table 6 presents the out-of-sample results when including the COVID-19 pandemic in the evaluation sample. The main findings established earlier remain broadly consistent, although some power is lost due to extreme outliers during the pandemic period. In the early weeks of the quarter, the NFCI remains more informative about the 10% quantile than the SPF one-step-ahead forecast. However, once the SPF nowcast becomes available, the SPF-GARCH model achieves the lowest losses. Similarly, for the 90% quantile and the predictive density of GDP growth, the SPF-GARCH model remains the significantly best-performing approach in most cases. As before, the GARCH model appears to utilize

the SPF median nowcasts more effectively than quantile regressions.

Table 6: Out-of-sample evaluation including the COVID pandemic

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.72	0.72	0.71	**0.34**	**0.34**	**0.34**	**0.34**	**0.34**
QR-NFCI	**0.62**	**0.64**	**0.64**	0.65	0.64	0.66	0.64	0.65
QR-SPF	0.74	0.74	0.73	0.39	0.37	0.37	0.37	0.37
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.53	0.53	**0.43**	**0.32**	**0.31**	**0.31**	**0.31**	**0.31**
QR-NFCI	0.62	0.63	0.67	0.66	0.66	0.66	0.66	0.67
QR-SPF	**0.53**	**0.53**	0.58	0.37	0.35	0.35	0.35	0.35
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	1.70	1.70	**1.62**	**1.07**	**1.05**	**1.05**	**1.05**	**1.05**
QR-NFCI	1.76	1.78	1.78	1.78	1.78	1.77	1.77	1.78
QR-SPF	**1.66**	**1.66**	1.65	1.13	1.09	1.09	1.09	1.09

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with consensus forecasts from the SPF in the conditional mean. The QR-SPF and QR-NFCI models refer to quantile regressions augmented with SPF consensus forecasts and the NFCI, respectively. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2024:Q4, yielding 140 quarterly observations.

6.3.2 The global financial crisis

Focusing on Growth-at-Risk during the GFC, the upper panel of Table 7 displays the real-time average 10% quantile losses from 2007:Q4 to 2009:Q3. The table does not contain information about the MCS due to the low statistical power resulting from the small number of observations (eight quarters).

At the beginning of the quarter, the QR-NFCI model provides the most accurate GaR nowcasts, suggesting that the NFCI is a strong predictor of economic downside risk during the financial crisis. For instance, in the third week of a quarter, the 10% quantile loss is less than one-third of the loss associated with the SPF-GARCH model. However, in the second half of the quarter, conditional on the SPF median nowcasts, the SPF-GARCH model and the QR-SPF model match the predictive performance of the QR-NFCI approach.

Figure 3 compares actual GDP against the median forecast of the SPF-GARCH in panel (a) and QR-NFCI in panel (b), respectively. The gray-shaded areas represent the

Table 7: Growth-at-Risk during economic turmoils

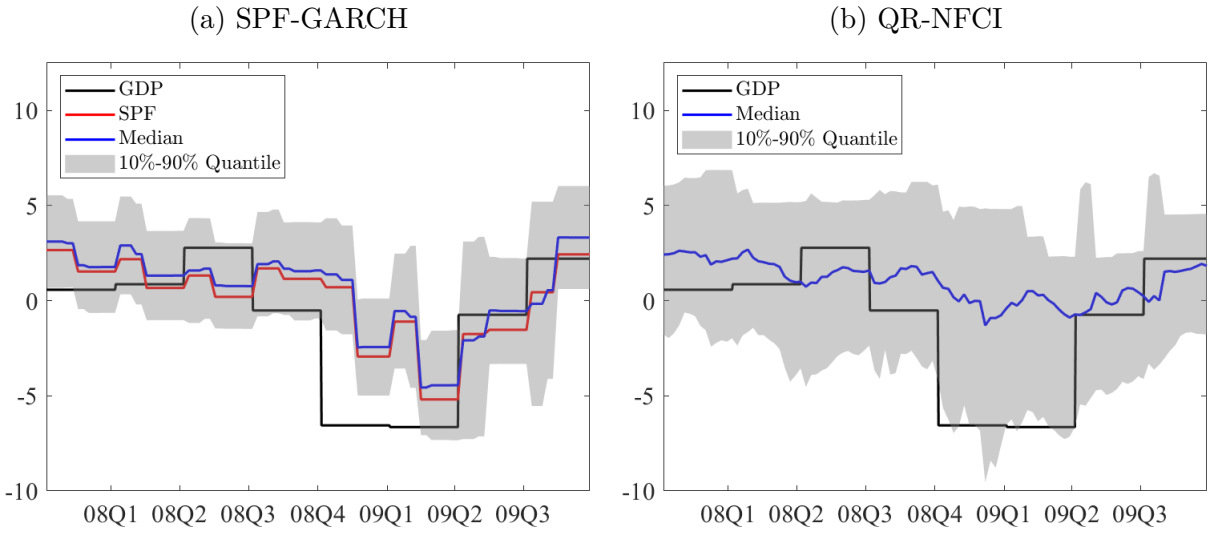
	week							
Financial Crisis	1	3	5	7	9	11	13	15
SPF-GARCH	1.16	1.16	1.01	0.33	0.34	0.34	0.33	0.33
QR-NFCI	**0.48**	**0.33**	**0.49**	0.32	**0.30**	0.34	**0.30**	0.45
QR-SPF	0.97	0.97	0.81	**0.30**	0.30	**0.30**	0.30	**0.30**
COVID-19	1	3	5	7	9	11	13	15
SPF-GARCH	4.53	4.53	4.53	**0.85**	**0.86**	**0.86**	**0.87**	**0.87**
QR-NFCI	4.04	4.08	4.14	4.22	4.29	4.23	4.10	4.07
QR-SPF	**3.88**	**3.88**	**3.81**	0.88	0.90	0.90	0.90	0.90

Notes: This table reports average losses for the 10% quantile score during the Global Financial Crisis and the COVID-19 pandemic. Columns indicate the week of the quarter when the nowcast is formed. Bold values highlight the lowest average loss within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with consensus forecasts from the SPF in the conditional mean. The QR-SPF and QR-NFCI models refer to quantile regressions augmented with SPF consensus forecasts and the NFCI, respectively. All models are estimated using an expanding window. The evaluation period in the top panel is 2007:Q4 to 2009:Q3 (8 observations), and in the bottom panel is 2020:Q1 to 2022:Q4 (12 observations).

predictive densities in terms of the 10% to 90% quantiles. Preceding the global financial crisis, quantile regressions with NFCI as conditioning information seem to provide overly pessimistic nowcasts of GaR. However, when GDP growth fell sharply in 2008:Q4, the conditional 10% quantiles of the QR-NFCI track the movements of GDP growth well. In contrast, even in the fourth quarter of 2008, the surveys panelists did not anticipate the severity of this event. Consequently, the SPF-GARCH model underestimates GaR in the fourth quarter of 2008. In the first quarter of 2009, the one-step-ahead median SPF forecast again resulted in a conditional 10% quantile that was well above the actual growth rate. Eventually, the economic assessment of the median SPF nowcast released in the middle of 2009:Q1 drops to -5.2%, and the associated 10%–90% quantile range covers the actual GDP growth rate of -6.4%.

In summary, at the onset of the GFC, the SPF-GARCH model suffers from less frequent updates, whereas the NFCI quickly reacts to new information. It takes the SPF median panelist until the second quarter of subsequently strongly negative growth rates to adjust accordingly. Nevertheless, starting from the middle of 2009:Q1, the SPF anticipates the magnitude and the timing of the financial crisis and the subsequent upswing well. This explains the relatively large 10% quantile losses of the SPF-GARCH in the first weeks of the quarter, and the mechanism of matching to the performance of the QR-NFCI nowcasts toward the end of the quarter.

Figure 3: Real-time nowcasts through the global financial crisis



Notes: This figure shows real-time nowcasts of quarterly US GDP growth during the global financial crisis from 2007:Q4 to 2009:Q3 for the SPF-GARCH model in panel (a) and the QR-NFCI approach in panel (b). The black line represents actual GDP growth, while the blue line shows the predicted median forecasts. The grey areas indicate density forecasts spanning the 10% to 90% conditional quantiles. In panel (a), the SPF-GARCH model assumes a normal distribution, so the median coincides with the mean, and the red line represents the SPF consensus forecasts.

6.3.3 The COVID-19 pandemic in isolation

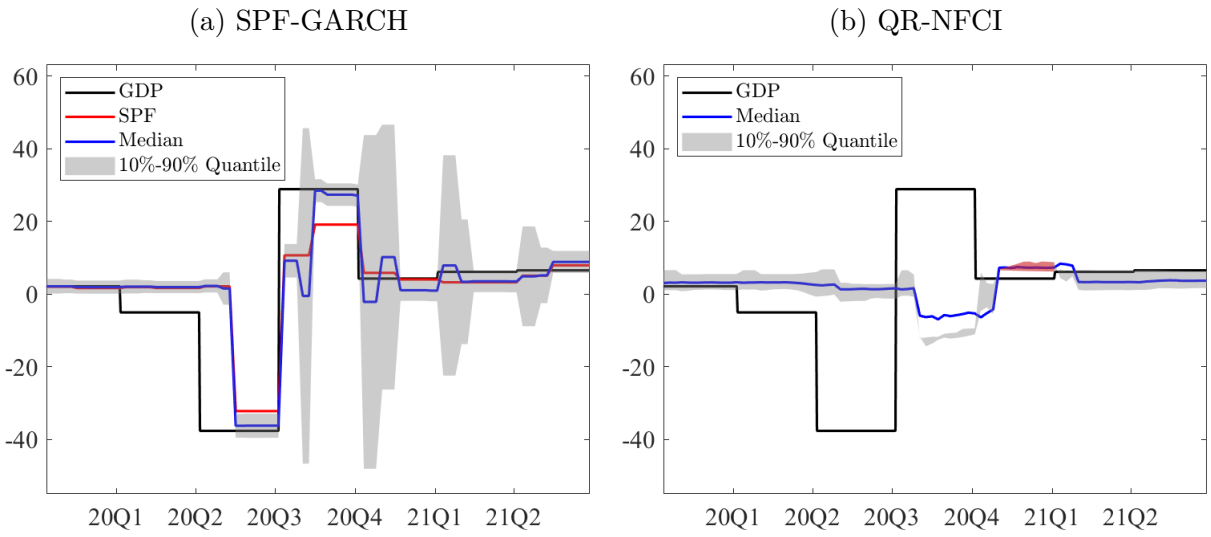
The nature of the shock to the economy represented by the COVID-19 pandemic is quite distinct from that of the global financial crisis. Thus, the bottom panel in Table 7 is concerned with the period 2020:Q1 to 2022:Q4. As becomes evident, the QR-NFCI nowcasts exhibit large 10% quantile losses. As panel (b) in Figure 4 shows, the QR-NFCI density nowcasts completely miss the down- and upswings of GDP growth during the COVID pandemic. Moreover, even extreme quantiles suffer from quantile crossing. Specifically, the median is not strictly between the estimated 10% and 90% quantile and during the fourth quarter of 2020, the 10% quantile lies above the 90% quantile, as indicated by the red-shaded area.[5]

In the first five weeks of a quarter, based on the SPF median one-step-ahead forecast, the SPF-GARCH model also exhibits considerable average 10% quantile losses. By week 7, with the release of the median nowcast, these losses decrease dramatically by over 80%. Panel (a) of Figure 4 illustrates how updating the SPF improves the GaR nowcasting accuracy. In the first quarter of 2020, at the onset of the lockdown measures, the SPF panelists did not adjust their assessment of the economic outlook. However, by mid-2020:Q2, the SPF released a median nowcast of -32.2% GDP growth, relatively close to

[5]Prior to the pandemic, there is no instance of the 10% and 90% quantiles crossing. Only in one quarter in the early nineties does the median lie above the 90% quantile.

the actual rate of -37.7%, aligning the predictive density closely with the actual outcome. Moreover, in the following quarter, the SPF median projections anticipated the timing and magnitude of the subsequent upswing.

Figure 4: Real-time nowcasts through the COVID-19 pandemic



Notes: This figure shows real-time nowcasts of quarterly US GDP growth during the COVID-19 pandemic from 2019:Q4 to 2021:Q2 for the SPF-GARCH model in panel (a) and the QR-NFCI approach in panel (b). The black line represents actual GDP growth, while the blue line shows the predicted median forecasts. The grey areas indicate density forecasts spanning the 10% to 90% conditional quantiles. In panel (a), the SPF-GARCH model assumes a normal distribution, so the median coincides with the mean, and the red line represents the SPF consensus forecasts. In panel (b), the red shading highlights periods where the predicted 10% quantile exceeds the 90% quantile.

In conclusion, while the SPF consensus one-step-ahead forecast in the second quarter of 2020 resulted in a large forecast error and wide prediction intervals in the subsequent quarters, the SPF consensus nowcasts improve the predictive density by shifting the center of the conditional distribution.[6]

6.4 GDP growth and skewness

The preferred model for nowcasting GDP growth quantiles and densities is the GARCH model with SPF nowcasts in the conditional mean, assuming normality. To assess whether this assumption limits predictive accuracy, I now allow for skewness and excess kurtosis by assuming a skewed t-distribution.[7] This extension is motivated by prior evidence

[6]From a practitioner's perspective, the large conditional variances in the quarters following the COVID-19 pandemic are undesirable. An easy fix is to treat the onset of the pandemic as an extreme outlier and replace the SPF one-step-ahead forecast by the median nowcast for 2020:Q2 in the subsequent quarters.

[7]Instead of assuming a parametric distribution, results not reported in the paper show that employing one-step-ahead filtered historical simulation, i.e., using the empirical distribution of the estimated standardized GARCH residuals, yields almost identical predictive densities.

suggesting time-varying asymmetry in the conditional GDP growth distribution during recessions (Adrian et al., 2019; Delle Monache et al., 2024), though some studies report little skewness (Brownlees and Souza, 2021; Carriero et al., 2022). I first compare predictive performance out-of-sample, then analyze the in-sample distributional shape using the approach of Adrian et al. (2019).

6.4.1 SPF-GARCH with skew *t*-distribution

In line with Adrian et al. (2019), the skew t-distribution used in this analysis is the one developed by Azzalini and Capitanio (2003). The probability density function of y, omitting subscripts for notational convenience, is given by

$$f(y;\mu,\sigma,\xi,\nu) = \frac{2}{\sigma} t\left(\frac{y-\mu}{\sigma};\nu\right) T\left(\xi\frac{y-\mu}{\sigma}\sqrt{\frac{\nu+1}{\nu+\frac{(y-\mu)^2}{\sigma^2}}};\nu+1\right), \tag{14}$$

where $t(\cdot)$ and $T(\cdot)$ denote the pdf and cdf of the Student t-distribution, respectively. The parameter μ governs location, σ controls scale, ξ determines skewness, and ν governs tail fatness. During the evaluation period, the conditional skew t-distribution of GDP growth for each week in each quarter is denoted by

$$y_{t|t,w} \sim \mathrm{St}(\mu_{t|t,w}, \sigma_{t|t,w}, \xi_{t|t,w}, \nu_{t|t,w}), \tag{15}$$

where $\mu_{t|t,w}$ and $\sigma_{t|t,w}$ are determined by equations (4) and (5), respectively. The log-likelihood is computed directly from the density in equation (14), and conditional quantiles are obtained numerically from the corresponding quantile function. The GARCH parameters and the skewness and tail-fatness parameters are estimated jointly.

As special cases, the density in equation (14) reduces to the Student t-distribution when $\xi = 0$, and to the skew normal when $\xi \neq 0$ and $\nu = \infty$. When $\xi = 0$ and $\nu = \infty$, it simplifies to the normal distribution with mean μ and variance σ^2.

Table 8 reports average losses for different SPF-GARCH specifications. 'Skewed t' refers to the unrestricted model allowing for both skewness and fat tails. 'Skewed Normal' and 'Symmetric t' are restricted variants allowing only for skewness or excess kurtosis, respectively. 'Normal' corresponds to the specification with normally distributed errors studied in Sections 6.1.

The results show limited gains from relaxing the normality assumption. Across the distribution of GDP growth, the unrestricted skewed t-distribution does not improve predictive accuracy. Only the skew normal specification yields small but significant improvements at the 90% quantile. This does not imply GDP growth is symmetric, but with short horizons and limited sample sizes, accounting for skewness or fat tails does not

Table 8: Out-of-sample evaluation of SPF-GARCH with skew t-distribution

	week							
10% quantile score	1	3	5	7	9	11	13	15
Normal	0.36	0.35	0.35	0.28	**0.29**	**0.29**	**0.29**	**0.29**
Skewed Normal	0.37	0.37	0.35	0.28	0.29	0.30	0.30	0.30
Symmetric t	**0.35**	**0.35**	**0.34**	**0.28**	0.29	0.29	0.29	0.29
Skewed t	0.36	0.36	0.34	0.28	0.30	0.30	0.30	0.30
90% quantile score	1	3	5	7	9	11	13	15
Normal	0.34	0.34	0.34	0.31	0.31	0.31	0.31	0.31
Skewed Normal	0.35	0.36	0.35	**0.31**	**0.30**	**0.30**	**0.30**	**0.30**
Symmetric t	**0.34**	**0.34**	**0.33**	0.31	0.31	0.31	0.31	0.31
Skewed t	0.35	0.35	0.34	0.32	0.31	0.31	0.31	0.31
CRPS	1	3	5	7	9	11	13	15
Normal	1.12	1.12	1.09	0.99	0.96	0.96	**0.96**	**0.96**
Skewed Normal	1.14	1.13	1.10	0.99	**0.95**	**0.96**	0.96	0.96
Symmetric t	**1.12**	**1.11**	**1.08**	0.99	0.96	0.96	0.96	0.96
Skewed t	1.14	1.14	1.09	**0.98**	0.97	0.97	0.97	0.97

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. All models are SPF-GARCH specifications; differences arise from the assumed distribution of the innovations. "Normal" and "Skewed Normal" denote Gaussian and skewed Gaussian innovations, while "Symmetric t" and "Skewed t" correspond to Student's t and skewed t-distributions. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

enhance forecast performance.

Figure A.2 shows that, conditional on SPF forecasts, the skew t-distribution captures notable variation in the mean (panel a–b) and some volatility dynamics (panel c–d). Similarly, panels (b) and (d) of Figure A.3 show that skewness is mostly centered around zero, while the kurtosis suggests moderately fat tails. In Figure A.4, panels (a) and (b) depict skewness for the skew normal distribution, while panels (c) and (d) illustrate kurtosis for the Student t-distribution. Once conditioned on the SPF nowcasts, the evidence again suggests a symmetric conditional GDP growth distribution with mild excess kurtosis.

6.4.2 A quantile matching approach

To further explore how conditioning information shapes the GDP growth distribution, this section applies the semi-parametric quantile matching approach of Adrian et al. (2019). First, I estimate quantile regressions of GDP growth on either the quarterly lag of the

NFCI or the SPF median nowcast for the 10%, 25%, 75%, and 90% quantiles. The analysis uses final vintage GDP data and covers the sample from 1973:Q1 to 2019:Q4.

Second, for each quarter, the parameters of the skew t-distribution are obtained by minimizing the squared distance between the estimated quantiles, denoted as $\hat{Q}^{\tau}_{t|x_{t-1}}$, where x_{t-1} is either the NFCI or the median SPF nowcast, and the corresponding quantiles from the skew t-distribution $F^{-1}(\tau, \mu_t, \sigma_t, \xi_t, \nu_t)$, based on the density in equation (14). The parameter estimates are obtained from:

$$\{\hat{\mu}_t, \hat{\sigma}_t, \hat{\xi}_t, \hat{\nu}_t\} = \underset{\mu,\sigma,\xi,\nu}{\arg\min} \sum_{\tau} \left(\hat{Q}^{\tau}_{t|x_{t-1}} - F^{-1}(\tau; \mu, \sigma, \xi, \nu)\right)^2, \tag{16}$$

where $\tau = \{0.1, 0.25, 0.75, 0.9\}$ represents the respective quantiles.

Figure A.5 illustrates the evolution of the implied mean and variance of the fitted skew t-distribution. When conditioning on the SPF, the implied mean exhibits stronger variation (panel a–b), while the NFCI leads to greater time-variation in the implied variance, especially during the 1970s, 1980s, and the global financial crisis (panel c–d), likely reflecting its inability to capture mean dynamics as effectively as the SPF.

Regarding higher moments, both conditioning variables imply some time-varying skewness (Figure A.6a–b), though the variation is more muted with the SPF. Kurtosis estimates (Figure A.6c–d) indicate modest fat tails but show no clear pattern linked to specific crises.

Overall, the in-sample evidence demonstrates that the shape of the skew t-distribution crucially depends on the conditioning information. The NFCI emphasizes variation in skewness and variance, while the SPF highlights shifts in the conditional mean, consistent with findings from Brownlees and Souza (2021) and Carriero et al. (2022). The conditional GDP growth distribution is thus mildly skewed and fat-tailed, helping to explain why relaxing the normality assumption yields limited out-of-sample gains.

6.5 Robustness analysis

6.5.1 Mean versus median SPF forecasts

Both the mean and median SPF forecasts can serve as measures of consensus. This paper uses the median, given its robustness to outliers. Table B.1 compares SPF-GARCH models using either the mean or median as conditioning information. For the 10% and 90% quantiles, differences in accuracy are small and insignificant, while the median yields slight but significant gains in density forecasts. Table B.2 shows similar results for quantile regressions. Overall, the findings are robust to the choice of consensus measure.

6.5.2 Forecaster disagreement

The SPF consensus forecasts derived from individual point predictions offer a useful summary of the economic outlook provided by the survey panelists. However, relying solely on a single measure of central tendency overlooks the rich information contained within the survey data.

One approach to capturing the cross-sectional dispersion in expectations among forecasters is to consider their *disagreement*. Abel et al. (2016) suggest the interquartile range (IQR) as a measure of forecasters' disagreement according to

$$SPF_{t,w}^{IQR} = SPF_{t,w}^{0.75} - SPF_{t,w}^{0.25},$$

where $SPF_{t,w}^{0.25}$ and $SPF_{t,w}^{0.75}$ denote the 25% and 75% quantiles of the panelists' point forecasts. The SPF also elicits probabilistic predictions through histogram forecasts, allowing for deriving measures of subjective *uncertainty*. However, the SPF asks respondents about fixed-event density forecasts, i.e., GDP growth for the current and the following calendar year. While disagreement is not a direct measure of uncertainty (see, for example, Abel et al., 2016; Glas, 2020), fixed-horizon disagreement is readily available from individual point predictions.

Table B.3 presents the out-of-sample results when incorporating the IQR into the SPF-GARCH and the QR-SPF model. Adding the IQR to the conditional variance improves Growth-at-Risk nowcasts and lowers the CRPS, though the gains are modest. Including the IQR in the conditional mean also improves performance at the 90% quantile, but improvements over the baseline SPF-GARCH model remain small.

The QR-SPF model with IQR is mostly included in the MCS. At the 90% quantile, conditional on one-step-ahead forecasts, adding disagreement yields the lowest losses. However, conditional on SPF nowcasts, the SPF-GARCH model performs better. Overall, disagreement offers only modest gains over models based solely on the median SPF forecast.

6.5.3 SPF recession probability

The SPF also reports the average perceived probability of a decline in real GDP over the current and next four quarters, known as the "Anxious Index" (Lahiri and Wang, 2013). Let $SPF_{t,w}^{\text{rec}}$ denote the one-step-ahead or nowcast of the recession probability in week w of quarter t.

Table B.4 reports the out-of-sample results for the SPF-GARCH and QR-SPF models when the recession measure is included. Conditional on one-step-ahead forecasts, the Anxious Index slightly improves downside risk predictions in the quantile regression model. However, the SPF-GARCH model continues to perform best and yields the lowest average

losses across the entire distribution.

6.5.4 Consensus forecast error distribution

This section exploits quantiles of historical forecast errors, following Becker et al. (2025). Specifically, the quantile nowcast is given by

$$\hat{Q}^{\text{error}}_{t|t,w}(\tau) = SPF_{t,w} + \hat{q}^{\tau}_{e^{SPF}_{t,w}}, \tag{17}$$

where $SPF_{t,w}$ is the median SPF one-step-ahead forecast or nowcast (depending on week w within quarter t) and $\hat{q}^{\tau}_{e^{SPF}_{t,w}}$ denotes the empirical τ-quantile of the real-time SPF median forecast error, $e^{SPF}_{t,w} \equiv y_{t,w} - SPF_{t,w}$.

Alternatively, Adams et al. (2021) propose modeling forecast error quantiles using quantile regressions on financial conditions. The conditional quantile is defined as

$$Q^{\text{NFCI}}_{e^{SPF}_{t,w}}(\tau) \equiv Q_{e^{SPF}_{t,w}}(\tau|x_{t,w}) = x_{t,w}\beta^{\tau}_{w},$$

where $x_{t,w}$ includes a constant and the real-time NFCI. The corresponding forecast is then

$$\hat{Q}^{\text{NFCI}}_{t|t,w}(\tau) = SPF_{t,w} + \hat{Q}^{\text{NFCI}}_{e^{SPF}_{t,w}}(\tau). \tag{18}$$

Table B.5 compares the out-of-sample accuracy of the SPF-GARCH model with the forecast-error-based approaches outlined in equations (17) and (18). The SPF-GARCH model delivers the best performance for downside risk. At the 90% quantile, forecasts based on $\hat{Q}^{\text{error}}_{t|t,w}(\tau)$ and $\hat{Q}^{\text{NFCI}}_{t|t,w}(\tau)$ yield slightly lower average losses than SPF-GARCH, but the differences are not statistically significant. Predictive densities are similarly accurate across all methods. Overall, SPF-GARCH remains the most reliable approach, particularly for forecasting Growth-at-Risk.

6.5.5 NFCI at higher frequency

Up to now, the NFCI has been analyzed at a quarterly frequency, using the latest available vintage within week w. To better exploit the high-frequency information, this section explores two alternative aggregation strategies. First, following Castelnuovo and Mori (2025), a monthly aggregation scheme is considered based on three monthly lags, $x_{t,w} = (NFCI_{t,w-1}, NFCI_{t,w-5}, NFCI_{t,w-9})$ estimated in real time and denoted by $NFCI^{m}_{t,w}$. Second, to assess the flow of weekly information more efficiently, a MIDAS-based approach is employed (Ghysels et al., 2004, 2007). The weekly NFCI is aggregated as

$NFCI^{\theta}_{t,w} = \sum_{i=1}^{m} \theta_i(\lambda_1, \lambda_2) NFCI_{t,w-i}$, where the unrestricted beta weights are defined as

$$\theta_i(\lambda_1, \lambda_2) = \frac{(i/(m+1))^{\lambda_1 - 1} (1 - i/(m+1))^{\lambda_2 - 1}}{\sum_{k=1}^{m} (i/(m+1))^{\lambda_1 - 1} (1 - i/(m+1))^{\lambda_2 - 1}}, \tag{19}$$

with $\lambda_1 \geq 1$, $\lambda_2 \geq 1$. The lag length is set to $m = 12$ to align with three months of weekly observations.

Table B.6 reports out-of-sample results for the SPF-GARCH model, both without NFCI and with NFCI, at quarterly, monthly ($NFCI^{m}_{t,w}$), and weekly MIDAS ($NFCI^{\theta}_{t,w}$) frequencies, incorporated into the conditional mean. Monthly aggregation yields modest nowcasting gains at the 10% quantile. However, once the SPF nowcast is available, incorporating higher-frequency NFCI into the mean equation does not meaningfully improve forecast accuracy. At the 90% quantile and for the CRPS, the baseline SPF-GARCH model without NFCI tends to perform best, though average differences in losses are small and statistically insignificant.

Similar insights are obtained when the NFCI is used as conditioning information in the volatility (Table B.7). In the QR-SPF model (Table B.8), neither monthly nor weekly NFCI aggregation improves GaR nowcasts. Overall, once SPF forecasts are incorporated, there is little evidence for applying MIDAS aggregation to the NFCI.

6.5.6 Rolling window estimation

To address concerns about asymptotically vanishing estimation uncertainty in recursive schemes (Giacomini and White, 2006), Table B.9 replicates the main results using a rolling window of 85 observations, corresponding to the number of SPF surveys available in the first quarter of 1990. The key findings remain robust. While increased estimation uncertainty reduces the power of the MCS at the 10% quantile, making the results nonsignificant, the SPF-GARCH model continues to yield the lowest losses once the nowcast becomes available. At the 90% quantile and for the CRPS, differences remain statistically significant.

6.5.7 Comparison with Bayesian Mixed-Frequency Models

To benchmark the SPF-GARCH model against data-rich methods, I compare its nowcasting performance to the Bayesian Mixed-Frequency Regression with Stochastic Volatility (BMF-SV) model analyzed in Carriero et al. (2022). The BMF-SV model integrates a broad set of macroeconomic and financial indicators at mixed frequencies while accounting for time-varying volatility.

One of the most accurate specification for GaR nowcasts in their study includes a comprehensive set of macroeconomic and financial indicators. The macroeconomic indica-

tors, available at monthly and weekly frequencies, include payroll employment, industrial production, real retail sales, housing starts, the Institute for Supply Management's manufacturing PMI, and initial unemployment insurance claims. The financial indicators consist of the NFCI, S&P 500 stock returns, the term spread between 10-year and 1-year Treasury yields, and the credit spread between Moody's Baa corporate yield and the 10-year Treasury yield.

Table B.10 compares the SPF-GARCH model to the BMF-SV model, using forecast data provided by Carriero et al. (2022). While the BMF-SV model achieves slightly lower losses at the 10% and 90% quantiles, the SPF-GARCH forecasts are almost always included in the MCS. Notably, both models deliver comparable performance by mid-quarter, once the SPF nowcasts become available.

These results underscore the efficiency of integrating survey expectations into a simple GARCH framework. Unlike the BMF-SV model, which requires processing a broad set of macroeconomic and financial indicators in real time, with substantial data and computational demands, the SPF-GARCH approach relies solely on consensus forecasts and is straightforward to update. Despite its simplicity and minimal data requirements, the SPF-GARCH model effectively captures time variation in both the lower and upper quantiles of the GDP growth distribution.

7 Conclusion

This study investigates the effectiveness of using SPF median projections for nowcasting quarterly GaR in the US. Nowcasts are constructed using both AR-GARCH-type models and quantile regressions. In a real-time out-of-sample forecasting evaluation, incorporating SPF consensus forecasts into the conditional mean of the GARCH model demonstrates superior predictive performance for GaR and the conditional density of GDP growth, especially following the release of the SPF nowcasts mid-quarter.

The SPF consensus projections are informative across the entire distribution of GDP growth, and the results based on the SPF-GARCH model, assuming normality, suggest notable time variation in both the lower and upper quantiles of GDP growth. Departing from symmetry, and incorporating fatness, does not enhance the forecasting accuracy for downside risk. While this does not imply that the conditional distribution of GDP growth is symmetric, these findings indicate that, given the short samples at hand, a more flexible distribution is not justified in the context of nowcasting GaR.

The SPF consensus forecasts emerge as highly informative for tail risk and the entire predictive distribution of GDP growth, particularly at short horizons. This study advocates for the inclusion of SPF consensus forecasts in the macroeconomic toolkit for

real-time assessment of downside risks. While the focus is on US GDP, future research could extend these findings to the ECB SPF as well as explore Inflation-at-Risk and Unemployment-at-Risk.

References

Abel, J., R. Rich, J. Song, and J. Tracy (2016). The measurement and behavior of uncertainty: Evidence from the ECB survey of professional forecasters. *Journal of Applied Econometrics 31*(3), 533–550.

Adams, P. A., T. Adrian, N. Boyarchenko, and D. Giannone (2021). Forecasting macroeconomic risks. *International Journal of Forecasting 37*(3), 1173–1191.

Adrian, T., N. Boyarchenko, and D. Giannone (2019, April). Vulnerable growth. *American Economic Review 109*(4), 1263–89.

Adrian, T., F. Grinberg, N. Liang, S. Malik, and J. Yu (2022, July). The term structure of Growth-at-Risk. *American Economic Journal: Macroeconomics 14*(3), 283–323.

Amburgey, A. J. and M. W. McCracken (2023). On the real-time predictive content of financial condition indices for growth. *Journal of Applied Econometrics 38*(2), 137–163.

Andrade, P., E. Ghysels, and J. Idier (2014). Inflation risk measures and their informational content. *Available at SSRN 2439607*.

Andreou, E., E. Ghysels, and A. Kourtellos (2013). Should macroeconomic forecasters use daily financial data and how? *Journal of Business & Economic Statistics 31*(2), 240–251.

Ang, A., G. Bekaert, and M. Wei (2007). Do macro variables, asset markets, or surveys forecast inflation better? *Journal of Monetary Economics 54*(4), 1163–1212.

Azzalini, A. and A. Capitanio (2003). Distributions generated by perturbation of symmetry with emphasis on a multivariate skew t-distribution. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 367–389.

Bańbura, M., D. Giannone, M. Modugno, and L. Reichlin (2013). Now-casting and the real-time data flow. In *Handbook of Economic Forecasting*, Volume 2, pp. 195–237. Elsevier.

Bańbura, M., F. Brenna, J. Paredes, and F. Ravazzolo (2021). Combining Bayesian VARs with survey density forecasts: Does it pay off?

Bassetti, F., R. Casarin, and M. Del Negro (2023). Inference on probabilistic surveys in macroeconomics with an application to the evolution of uncertainty in the survey of professional forecasters during the covid pandemic. In *Handbook of Economic Expectations*, pp. 443–476. Academic Press.

Becker, F., F. Krüger, and M. Schienle (2025). Simple macroeconomic forecast distributions for the G7 economies. *The Annals of Applied Statistics 19*(4), 2878–2897.

Brave, S. and R. A. Butters (2012, June). Diagnosing the financial system: Financial conditions and financial stress. *International Journal of Central Banking 8*(2), 191–239.

Brownlees, C. and A. B. Souza (2021). Backtesting global growth-at-risk. *Journal of Monetary Economics 118*, 312–330.

Carriero, A., T. E. Clark, and M. Marcellino (2022). Nowcasting tail risk to economic activity at a weekly frequency. *Journal of Applied Econometrics 37*(5), 843–866.

Carriero, A., T. E. Clark, and M. Marcellino (2024). Capturing macro-economic tail risks with bayesian vector autoregressions. *Journal of Money, Credit and Banking 56*(5), 1099–1127.

Castelnuovo, E. and L. Mori (2025). Uncertainty, skewness, and the business cycle through the MIDAS lens. *Journal of Applied Econometrics 40*(1), 89–107.

Clark, T. E., M. W. McCracken, and E. Mertens (2020). Modeling time-varying uncertainty of multiple-horizon forecast errors. *The Review of Economics and Statistics 102*(1), 17–33.

Clements, M. P. (2010). Explanations of the inconsistencies in survey respondents' forecasts. *European Economic Review 54*(4), 536–549.

Clements, M. P. (2014). Probability distributions or point predictions? Survey forecasts of US output growth and inflation. *International Journal of Forecasting 30*(1), 99–117.

Clements, M. P., R. W. Rich, and J. S. Tracy (2023). Surveys of professionals. In *Handbook of Economic Expectations*, pp. 71–106. Academic Press.

De Nicolò, G. and M. Lucchetta (2017). Forecasting tail risks. *Journal of Applied Econometrics 32*(1), 159–170.

Delle Monache, D., A. De Polis, and I. Petrella (2024). Modeling and forecasting macroeconomic downside risk. *Journal of Business & Economic Statistics 42*(3), 1010–1025.

Diebold, F. X. and R. S. Mariano (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics 13*(3), 253–263.

Elliott, G., I. Komunjer, and A. Timmermann (2008). Biases in macroeconomic forecasts: Irrationality or asymmetric loss? *Journal of the European Economic Association 6*(1), 122–157.

Fagiolo, G., M. Napoletano, and A. Roventini (2008). Are output growth-rate distributions fat-tailed? Some evidence from OECD countries. *Journal of Applied Econometrics 23*(5), 639–669.

Faust, J. and J. H. Wright (2013). Forecasting inflation. In *Handbook of Economic Forecasting*, Volume 2, pp. 2–56. Elsevier.

Ferrara, L., M. Mogliani, and J.-G. Sahuc (2022). High-frequency monitoring of growth at risk. *International Journal of Forecasting 38*(2), 582–595.

Figueres, J. M. and M. Jarociński (2020). Vulnerable growth in the euro area: Measuring the financial conditions. *Economics Letters 191*, 109126.

Ganics, G., B. Rossi, and T. Sekhposyan (2024). From fixed-event to fixed-horizon density forecasts: Obtaining measures of multihorizon uncertainty from survey density forecasts. *Journal of Money, Credit and Banking 56*(7), 1675–1704.

Ghysels, E., P. Santa-Clara, and R. Valkanov (2004). The MIDAS touch: Mixed data sampling regression models. Cirano working papers, CIRANO.

Ghysels, E., A. Sinko, and R. Valkanov (2007). MIDAS regressions: Further results and new directions. *Econometric Reviews 26*(1), 53–90.

Giacomini, R. and H. White (2006). Tests of conditional predictive ability. *Econometrica 74*(6), 1545–1578.

Giannone, D., L. Reichlin, and D. Small (2008). Nowcasting: The real-time informational content of macroeconomic data. *Journal of Monetary Economics 55*(4), 665–676.

Giglio, S., B. Kelly, and S. Pruitt (2016). Systemic risk and the macroeconomy: An empirical evaluation. *Journal of Financial Economics 119*(3), 457–471.

Glas, A. (2020). Five dimensions of the uncertainty–disagreement linkage. *International Journal of Forecasting 36*(2), 607–627.

Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association 106*(494), 746–762.

Gneiting, T. and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association 102*(477), 359–378.

Hansen, P. R., A. Lunde, and J. M. Nason (2011). The model confidence set. *Econometrica 79*(2), 453–497.

Koenker, R. and G. Bassett (1978). Regression quantiles. *Econometrica 46*(1), 33–50.

Krüger, F., T. E. Clark, and F. Ravazzolo (2017). Using entropic tilting to combine BVAR forecasts with external nowcasts. *Journal of Business & Economic Statistics 35*(3), 470–485.

Krüger, F. and H. Plett (2024). Prediction intervals for economic fixed-event forecasts. *The Annals of Applied Statistics 18*(3), 2635 – 2655.

Lahiri, K. and J. G. Wang (2013). Evaluating probability forecasts for GDP declines using alternative methodologies. *International Journal of Forecasting 29*(1), 175–190.

Mincer, J. A. and V. Zarnowitz (1969, December). The evaluation of economic forecasts. In *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER Chapters, pp. 3–46. National Bureau of Economic Research, Inc.

Plagborg-Møller, M., L. Reichlin, G. Ricco, and T. Hasenzagl (2020). When is growth at risk? *Brookings Papers on Economic Activity 2020*(1), 167–229.

Prasad, M. A., S. Elekdag, M. P. Jeasakul, R. Lafarguette, M. A. Alter, A. X. Feng, and C. Wang (2019). *Growth at risk: Concept and application in IMF country surveillance*. International Monetary Fund.

Reichlin, L., G. Ricco, and T. Hasenzagl (2020). Financial variables as predictors of real growth vulnerability.

Reifschneider, D. and P. Tulip (2019). Gauging the uncertainty of the economic outlook using historical forecasting errors: The Federal Reserve's approach. *International Journal of Forecasting 35*(4), 1564–1582.

Rossi, B. and T. Sekhposyan (2019). Alternative tests for correct specification of conditional predictive densities. *Journal of Econometrics 208*(2), 638–657.

Sheppard, K. (2009). MFE MATLAB function reference financial econometrics. *URL https://www.kevinsheppard.com/files/code/matlab/mfe-toolbox-documentation.pdf*.

Stark, T. (2010). Realistic evaluation of real-time forecasts in the survey of professional forecasters. *Research Rap Special Report* (May).

Wallis, K. F. (1986). Forecasting with an econometric model: The 'ragged edge' problem. *Journal of Forecasting 5*(1), 1–13.

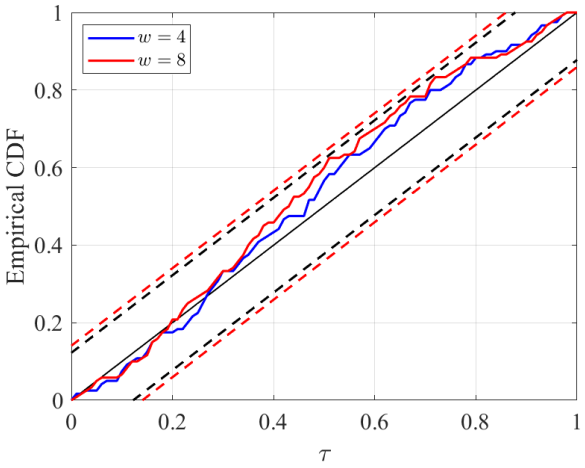
Zarnowitz, V. (1985). Rational expectations and macroeconomic forecasts. *Journal of Business & Economic Statistics 3*(4), 293–311.

Appendices

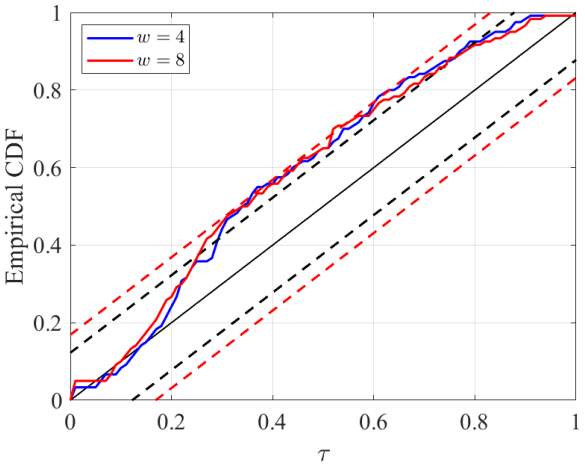
A Figures

Figure A.1: Probability integral transform

(a) SPF-GARCH

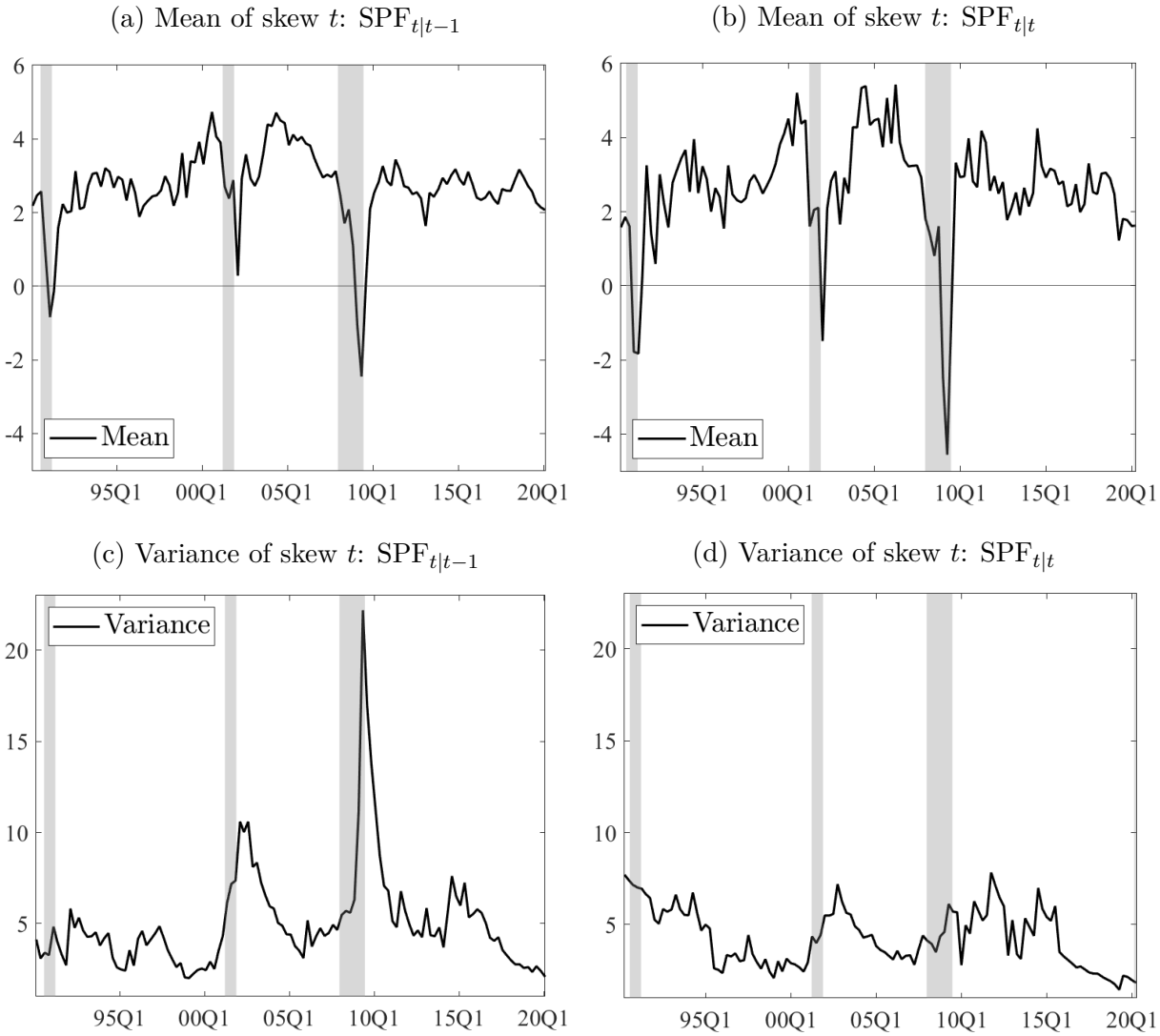


(b) QR-NFCI



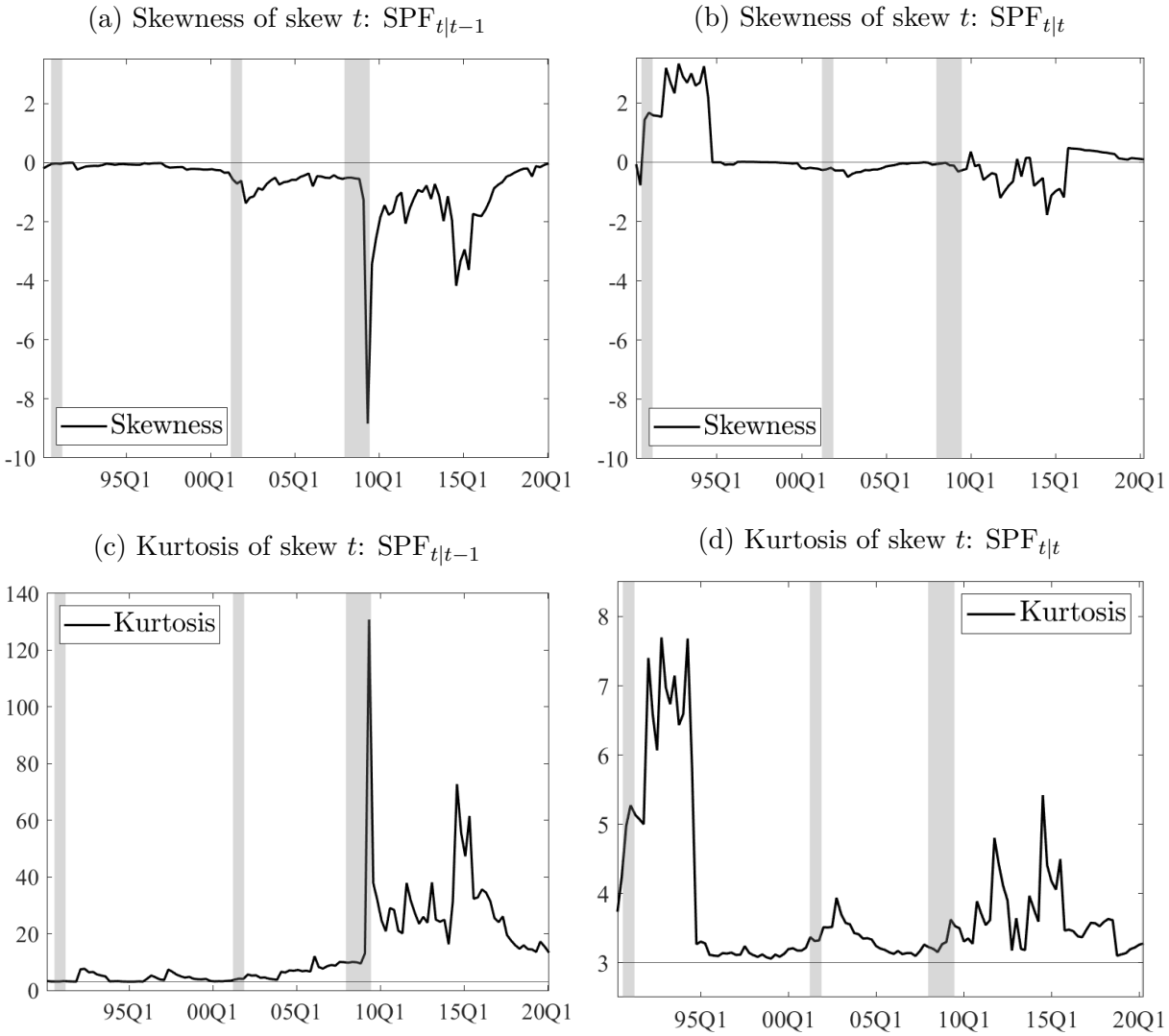
Notes: This figure shows the probability integral transform (PIT) over the out-of-sample evaluation period from 1990:Q1 to 2019:Q4 for the SPF-GARCH model in panel (a) and the QR-NFCI approach in panel (b). The black dashed lines indicate 95% confidence bands under the null of uniformity and independence of the PIT. The red dashed lines represent 95% bootstrap confidence bands constructed under the null of uniformity only, following Rossi and Sekhposyan (2019).

Figure A.2: Mean and variance of the skew t-distribution



Notes: This figure illustrates real-time estimates of the conditional mean and conditional variance of the skew t-distribution over the out-of-sample evaluation period from 1990:Q1 to 2019:Q4 for the SPF-GARCH model. Panels (a) and (b) display the conditional mean, while panels (c) and (d) show the conditional variance. Panels (a) and (c) utilize the SPF one-step-ahead median forecasts, whereas panels (b) and (d) employ the nowcasts. The shaded areas indicate US recessions as identified by the GDP-based recession indicator.

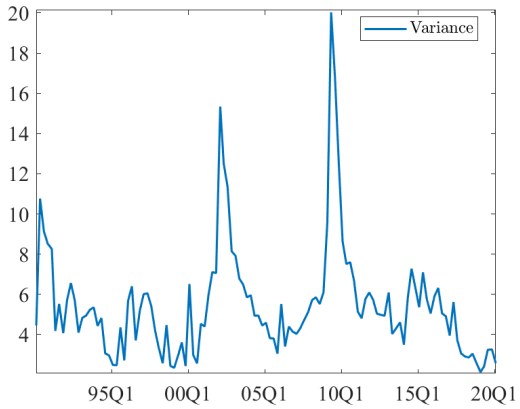
Figure A.3: Skewness and kurtosis of the skew t-distribution



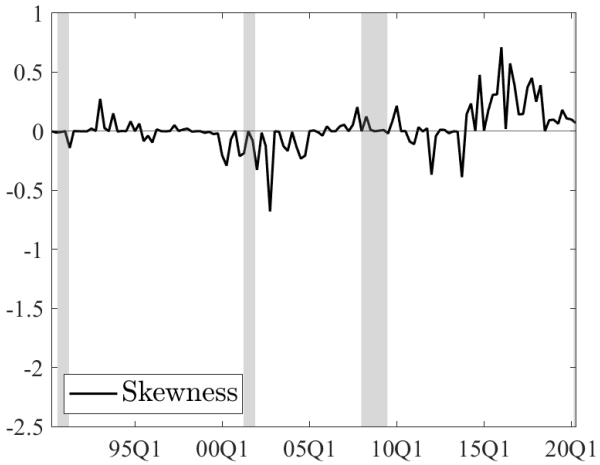
Notes: This figure illustrates real-time estimates of the conditional skewness and conditional kurtosis of the skew t-distribution over the out-of-sample evaluation period from 1990:Q1 to 2019:Q4 for the SPF-GARCH model. Panels (a) and (b) display the conditional skewness, while panels (c) and (d) show the conditional kurtosis. Panels (a) and (c) utilize the SPF one-step-ahead median forecasts, whereas panels (b) and (d) employ the nowcasts. The shaded areas indicate US recessions as identified by the GDP-based recession indicator.

Figure A.4: Skewness of the skew normal and kurtosis of the t-distribution

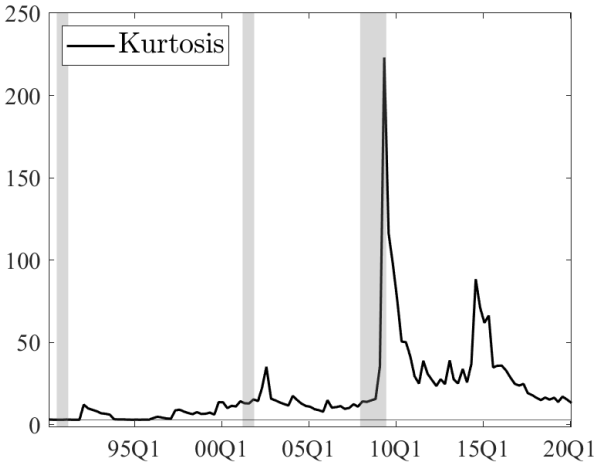
(a) Skew normal distribution: $\text{SPF}_{t|t-1}$



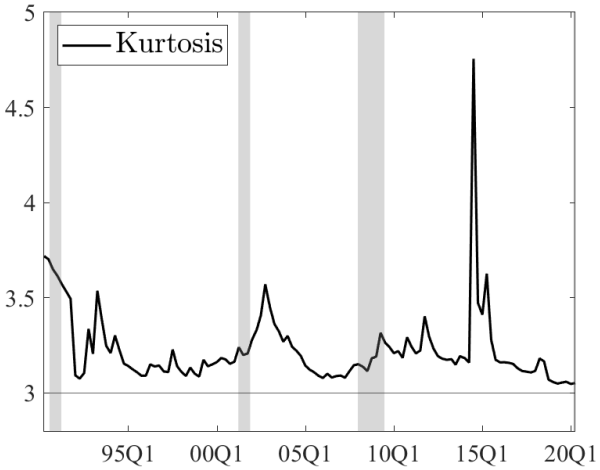
(b) Skew normal distribution: $\text{SPF}_{t|t}$



(c) Student t-distribution: $\text{SPF}_{t|t-1}$

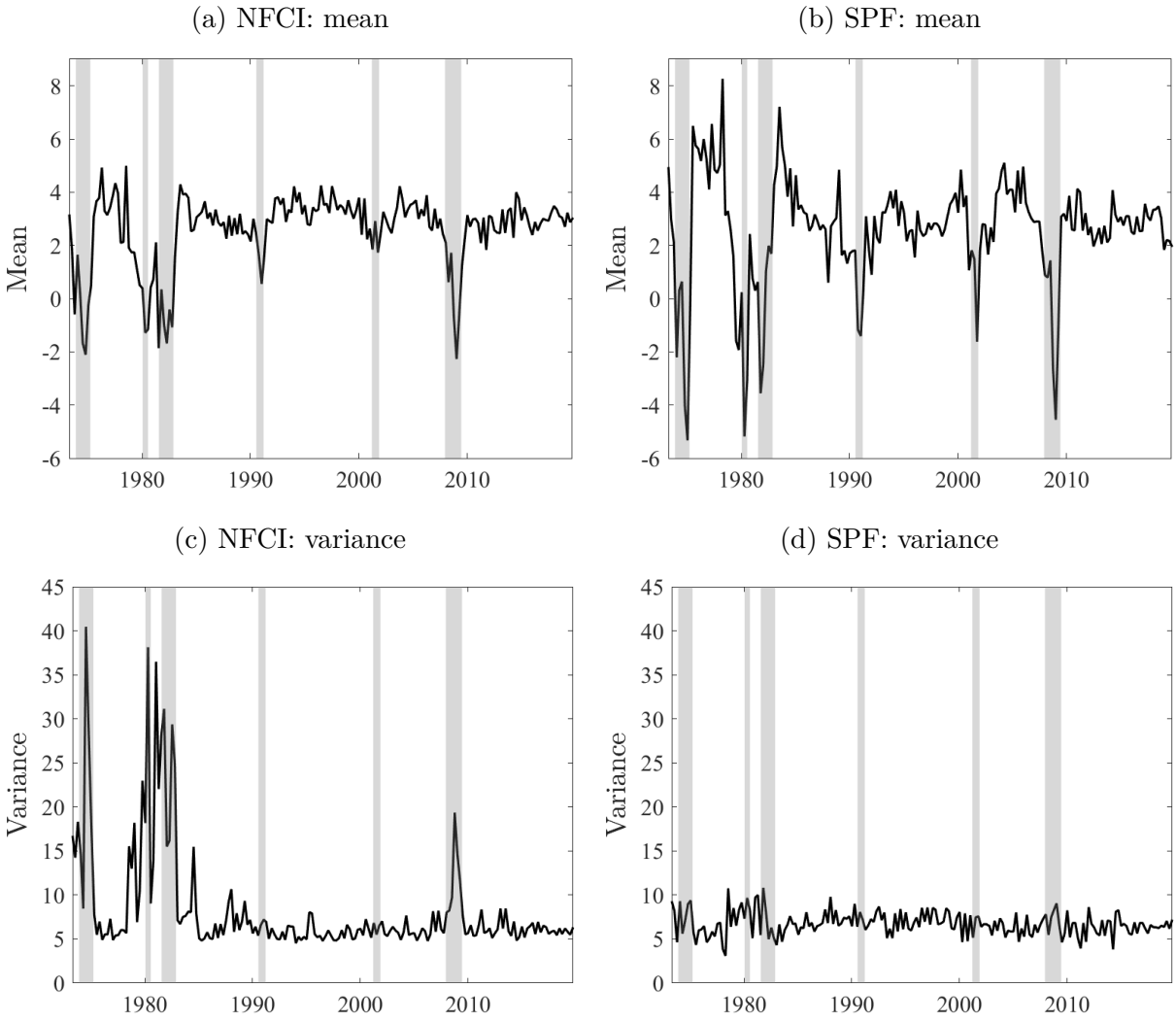


(d) Student t-distribution: $\text{SPF}_{t|t}$



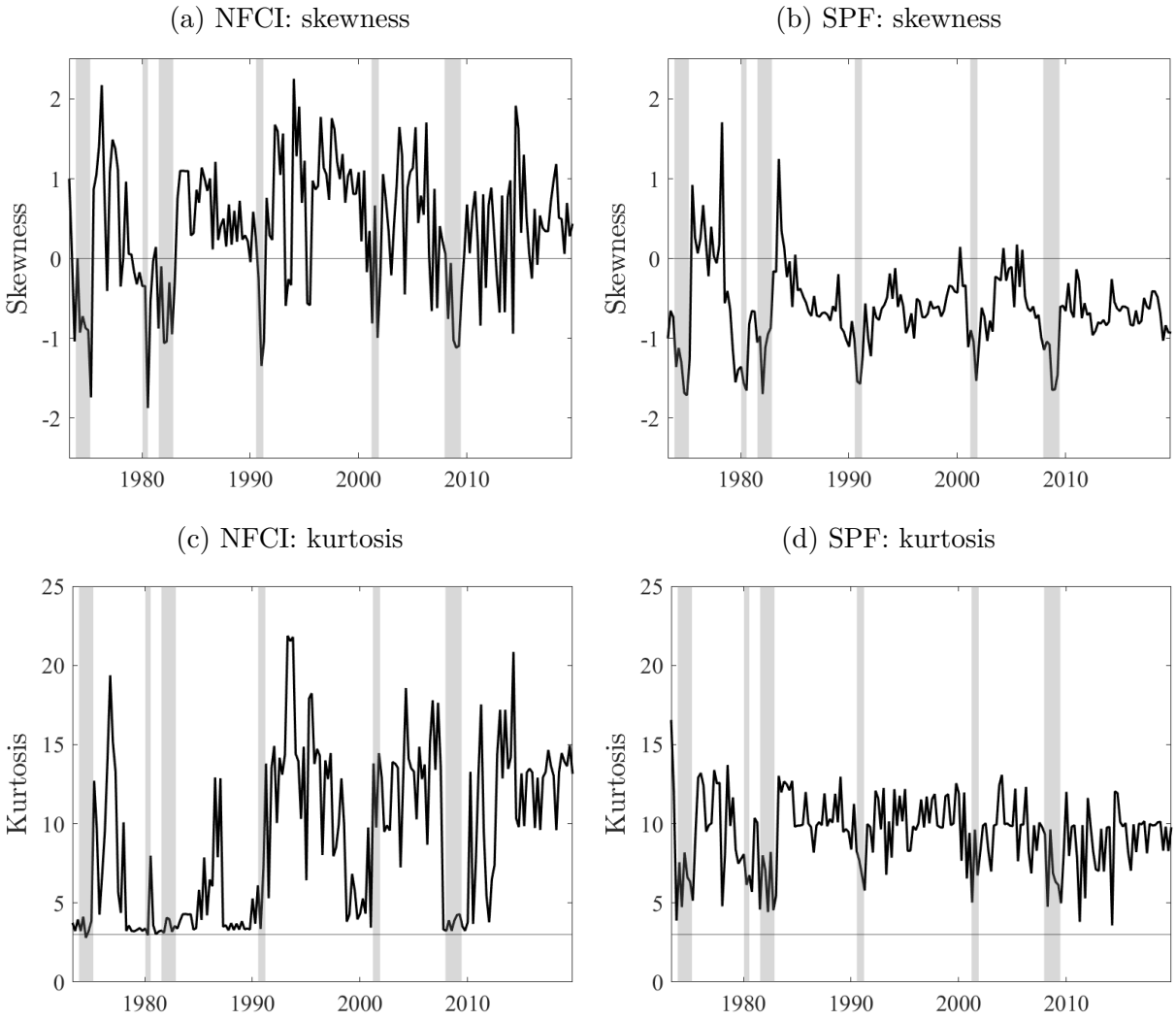
Notes: This figure illustrates real-time estimates of the conditional skewness of the skew normal distribution and conditional kurtosis of the Student t-distribution over the out-of-sample evaluation period from 1990:Q1 to 2019:Q4 for the SPF-GARCH model. Panels (a) and (b) display the conditional skewness, while panels (c) and (d) show the conditional kurtosis. Panels (a) and (c) utilize the SPF one-step-ahead median forecasts, whereas panels (b) and (d) employ the nowcasts. The shaded areas indicate US recessions as identified by the GDP-based recession indicator.

Figure A.5: In-sample mean and variance



Notes: This figure illustrates the in-sample mean and variance of GDP growth from 1973:Q1 to 2019:Q4, estimated using the semi-parametric quantile-matching approach described in Section 6.4.2. As conditioning information, panels (a) and (c) use the latest vintage of the NFCI, while panels (b) and (d) employ the SPF median nowcasts. The shaded areas indicate US recessions as identified by the GDP-based recession indicator.

Figure A.6: In-sample skewness and kurtosis



Notes: This figure illustrates the in-sample skewness and kurtosis of GDP growth from 1973:Q1 to 2019:Q4, estimated using the semi-parametric quantile-matching approach described in Section 6.4.2. As conditioning information, panels (a) and (c) use the latest vintage of the NFCI, while panels (b) and (d) employ the SPF median nowcasts. The shaded areas indicate US recessions as identified by the GDP-based recession indicator.

B Tables

Table B.1: SPF-GARCH using median versus mean SPF

	week							
10% quantile score	1	3	5	7	9	11	13	15
Mean SPF	0.36	0.35	0.35	0.28	**0.29**	**0.29**	**0.29**	**0.29**
Median SPF	**0.36**	**0.35**	**0.35**	**0.28**	0.29	0.29	0.29	0.29
90% quantile score	1	3	5	7	9	11	13	15
Mean SPF	0.34	0.34	0.34	0.32	0.31	0.31	0.31	0.31
Median SPF	**0.34**	**0.34**	**0.34**	**0.31**	**0.31**	**0.31**	**0.31**	**0.31**
CRPS	1	3	5	7	9	11	13	15
Mean SPF	1.13	1.12	1.10	1.01	0.98	0.98	0.98	0.98
Median SPF	**1.12**	**1.12**	**1.09**	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with forecasts from the SPF, where the rows labeled "Mean SPF" and "Median SPF" correspond to using the SPF consensus mean and median forecasts, respectively. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.2: QR-SPF using median versus mean SPF

	week							
10% quantile score	1	3	5	7	9	11	13	15
Mean SPF	**0.43**	**0.43**	**0.42**	**0.34**	**0.31**	**0.31**	**0.31**	**0.31**
Median SPF	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
90% quantile score	1	3	5	7	9	11	13	15
Mean SPF	0.39	0.39	0.39	0.36	0.34	0.34	0.34	0.34
Median SPF	**0.39**	**0.39**	**0.38**	**0.36**	**0.33**	**0.33**	**0.33**	**0.33**
Mean SPF	1.18	1.18	1.15	1.05	1.01	1.01	1.01	1.01
Median SPF	**1.17**	**1.17**	**1.14**	**1.03**	**1.00**	**1.00**	**1.00**	**1.00**

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The QR-SPF model refers to a quantile regression including forecasts from the SPF, where the rows labeled "Mean SPF" and "Median SPF" correspond to using the SPF consensus mean and median forecasts, respectively. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.3: Out-of-sample evaluation using the SPF interquantile range

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	0.28	0.29	0.29	0.29	0.29
$+x^{\mu}_{t,w} = SPF^{IQR}_{t,w}$	0.36	0.36	0.35	0.28	0.29	0.29	0.29	0.29
$+x^{\sigma}_{t,w} = SPF^{IQR}_{t,w}$	**0.35**	**0.35**	**0.34**	**0.27**	**0.28**	**0.28**	**0.28**	**0.28**
QR-SPF	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
$+SPF^{IQR}_{t,w}$	0.37	0.37	0.36	0.31	0.28	0.29	0.29	0.29
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.34	0.34	0.34	0.31	0.31	0.31	0.31	0.31
$+x^{\mu}_{t,w} = SPF^{IQR}_{t,w}$	0.33	0.33	0.33	**0.31**	**0.30**	**0.30**	**0.30**	**0.30**
$+x^{\sigma}_{t,w} = SPF^{IQR}_{t,w}$	0.35	0.35	0.34	0.32	0.31	0.31	0.31	0.31
QR-SPF	0.39	0.39	0.38	0.36	0.33	0.33	0.33	0.33
$+SPF^{IQR}_{t,w}$	**0.33**	**0.32**	**0.29**	0.31	0.32	0.32	0.32	0.32
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	1.12	1.12	1.09	0.99	0.96	0.96	0.96	0.96
$+x^{\mu}_{t,w} = SPF^{IQR}_{t,w}$	1.12	1.12	1.08	0.99	0.95	0.96	0.95	0.95
$+x^{\sigma}_{t,w} = SPF^{IQR}_{t,w}$	**1.11**	**1.11**	1.09	**0.99**	**0.95**	**0.96**	**0.95**	**0.95**
QR-SPF	1.17	1.17	1.14	1.03	1.00	1.00	1.00	1.00
$+SPF^{IQR}_{t,w}$	1.13	1.13	**1.08**	1.00	0.97	0.97	0.97	0.97

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with SPF consensus forecasts in the conditional mean, while the QR-SPF model refers to a quantile regression including SPF consensus forecasts. $SPF^{IQR}_{t,w}$ denotes the cross-sectional interquartile range among the forecasters. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.4: Out-of-sample evaluation using the SPF recession probability

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	0.28	**0.29**	**0.29**	**0.29**	**0.29**
$+x^{\mu}_{t,w} = SPF^{\text{rec}}_{t,w}$	0.35	0.35	0.34	**0.28**	0.29	0.29	0.29	0.29
$+x^{\sigma}_{t,w} = SPF^{\text{rec}}_{t,w}$	0.38	0.38	0.37	0.31	0.30	0.30	0.30	0.30
QR-SPF	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
$+SPF^{\text{rec}}_{t,w}$	**0.33**	**0.33**	**0.33**	0.29	0.29	0.29	0.29	0.29
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	**0.34**	**0.34**	**0.34**	**0.31**	**0.31**	**0.31**	**0.31**	**0.31**
$+x^{\mu}_{t,w} = SPF^{\text{rec}}_{t,w}$	0.34	0.34	0.34	0.32	0.32	0.31	0.31	0.31
$+x^{\sigma}_{t,w} = SPF^{\text{rec}}_{t,w}$	0.37	0.37	0.36	0.34	0.32	0.31	0.32	0.32
QR-SPF	0.39	0.39	0.38	0.36	0.33	0.33	0.33	0.33
$+SPF^{\text{rec}}_{t,w}$	0.38	0.38	0.36	0.34	0.33	0.33	0.33	0.33
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	**1.12**	1.12	1.09	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
$+x^{\mu}_{t,w} = SPF^{\text{rec}}_{t,w}$	1.12	**1.11**	**1.09**	0.99	0.97	0.97	0.97	0.97
$+x^{\sigma}_{t,w} = SPF^{\text{rec}}_{t,w}$	1.15	1.15	1.12	1.01	0.97	0.97	0.97	0.97
QR-SPF	1.17	1.17	1.14	1.03	1.00	1.00	1.00	1.00
$+SPF^{\text{rec}}_{t,w}$	1.14	1.14	1.11	1.01	0.98	0.99	0.98	0.98

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with SPF consensus forecasts in the conditional mean, while the QR-SPF model refers to a quantile regression including SPF consensus forecasts. $SPF^{\text{rec}}_{t,w}$ denotes the recession probability from the SPF. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.5: Out-of-sample evaluation using SPF forecast errors

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	**0.36**	**0.35**	**0.35**	**0.28**	**0.29**	**0.29**	**0.29**	**0.29**
$\hat{Q}^{\text{error}}_{t\|t,w}(0.1)$	0.43	0.43	0.43	0.38	0.38	0.38	0.38	0.38
$\hat{Q}^{\text{NFCI}}_{t\|t,w}(0.1)$	0.36	0.36	0.35	0.34	0.36	0.36	0.36	0.38
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	**0.34**	**0.34**	**0.34**	**0.31**	0.31	0.31	0.31	0.31
$\hat{Q}^{\text{error}}_{t\|t,w}(0.9)$	0.40	0.40	0.40	0.34	0.30	0.30	0.30	0.30
$\hat{Q}^{\text{NFCI}}_{t\|t,w}(0.9)$	0.35	0.34	0.34	0.31	**0.29**	**0.28**	**0.28**	**0.28**
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	1.12	1.12	**1.09**	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
$\hat{Q}^{\text{error}}_{t\|t,w}(\tau)$	1.13	1.13	1.13	1.04	1.03	1.03	1.03	1.03
$\hat{Q}^{\text{NFCI}}_{t\|t,w}(\tau)$	**1.09**	**1.09**	1.09	1.02	1.03	1.04	1.04	1.05

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with SPF consensus forecasts in the conditional mean. $\hat{Q}^{\text{error}}t|t,w(0.1)$ and $\hat{Q}^{\text{NFCI}}t|t,w(0.1)$ correspond to the forecast specifications defined in equations (17) and (18), respectively. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.6: Out-of-sample evaluation of MIDAS-NFCI in the conditional mean

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	**0.28**	0.29	0.29	0.29	0.29
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.37	0.35	0.34	0.28	**0.27**	**0.27**	**0.27**	**0.27**
$+x^{\mu}_{t,w} = NFCI^{m}_{t,w}$	**0.35**	**0.33**	**0.33**	0.30	0.30	0.30	0.31	0.30
$+x^{\mu}_{t,w} = NFCI^{\theta}_{t,w}$	0.38	0.36	0.35	0.29	0.27	0.27	0.27	0.27
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.34	**0.34**	0.34	**0.31**	**0.31**	**0.31**	**0.31**	**0.31**
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.37	0.36	0.34	0.32	0.31	0.31	0.31	0.31
$+x^{\mu}_{t,w} = NFCI^{m}_{t,w}$	**0.34**	0.35	**0.32**	0.33	0.32	0.32	0.33	0.32
$+x^{\mu}_{t,w} = NFCI^{\theta}_{t,w}$	0.38	0.38	0.36	0.33	0.32	0.31	0.31	0.32
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	**1.12**	**1.12**	1.09	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
$+x^{\mu}_{t,w} = NFCI_{t,w}$	1.18	1.16	1.09	1.01	0.97	0.97	0.97	0.98
$+x^{\mu}_{t,w} = NFCI^{m}_{t,w}$	1.14	1.14	**1.07**	1.04	1.01	1.00	1.02	1.01
$+x^{\mu}_{t,w} = NFCI^{\theta}_{t,w}$	1.20	1.21	1.14	1.01	0.98	0.98	0.97	0.98

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with SPF consensus forecasts in the conditional mean. $NFCI_{t,w}$ adds the weekly NFCI, $NFCI^{m}_{t,w}$ denotes inclusion of three monthly lags, and $NFCI^{\theta}_{t,w}$ denotes a MIDAS specification of the NFCI, each in the conditional mean. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.7: Out-of-sample evaluation of MIDAS-NFCI in the conditional volatility

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	0.28	0.29	0.29	0.29	0.29
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	**0.35**	**0.34**	**0.33**	**0.28**	**0.28**	**0.28**	**0.28**	0.28
$+x^{\sigma}_{t,w} = NFCI^{m}_{t,w}$	0.60	1.07	0.42	0.32	0.28	0.28	0.29	0.31
$+x^{\sigma}_{t,w} = NFCI^{\theta}_{t,w}$	0.36	0.35	0.34	0.28	0.28	0.28	0.28	**0.27**
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	**0.34**	**0.34**	**0.34**	**0.31**	**0.31**	0.31	**0.31**	**0.31**
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.37	0.36	0.36	0.34	0.32	0.32	0.32	0.31
$+x^{\sigma}_{t,w} = NFCI^{m}_{t,w}$	0.60	1.06	0.41	0.34	0.32	**0.31**	0.32	0.32
$+x^{\sigma}_{t,w} = NFCI^{\theta}_{t,w}$	0.36	0.37	0.36	0.34	0.32	0.32	0.32	0.32
CRPS	1	3	5	7	9	11	13	15
SPF-GARCH	**1.12**	**1.12**	**1.09**	**0.99**	**0.96**	**0.96**	**0.96**	**0.96**
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	1.14	1.14	1.10	1.01	0.98	0.98	0.98	0.98
$+x^{\sigma}_{t,w} = NFCI^{m}_{t,w}$	1.54	2.42	1.20	1.04	0.99	0.98	0.99	1.00
$+x^{\sigma}_{t,w} = NFCI^{\theta}_{t,w}$	1.16	1.16	1.11	1.02	0.98	0.98	0.98	0.98

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with SPF consensus forecasts in the conditional volatility. $NFCI_{t,w}$ adds the weekly NFCI, $NFCI^{m}_{t,w}$ denotes inclusion of three monthly lags, and $NFCI^{\theta}_{t,w}$ denotes a MIDAS specification of the NFCI, each in the conditional mean. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.8: Out-of-sample evaluation of quntile regression with MIDAS-NFCI

	week							
10% quantile score	1	3	5	7	9	11	13	15
QR-SPF	0.44	0.44	0.43	0.35	0.31	0.31	0.31	0.31
$+NFCI_{t,w}$	**0.29**	**0.30**	**0.29**	**0.28**	**0.27**	**0.27**	0.28	0.28
$+NFCI^{m}_{t,w}$	0.32	0.30	0.33	0.30	0.28	0.29	0.30	**0.27**
$+NFCI^{\theta}_{t,w}$	0.36	0.34	0.30	0.28	0.27	0.28	**0.28**	0.28
90% quantile score	1	3	5	7	9	11	13	15
QR-SPF	0.39	0.39	0.38	0.36	**0.33**	**0.33**	0.33	**0.33**
$+NFCI_{t,w}$	0.41	0.39	0.36	0.35	0.34	0.34	0.34	0.34
$+NFCI^{m}_{t,w}$	0.40	0.39	**0.35**	**0.35**	0.34	0.33	**0.31**	0.33
$+NFCI^{\theta}_{t,w}$	**0.37**	**0.38**	0.36	0.35	0.34	0.34	0.33	0.34
CRPS	1	3	5	7	9	11	13	15
QR-SPF	1.17	1.17	1.14	1.03	1.00	1.00	1.00	1.00
$+NFCI_{t,w}$	**1.17**	1.17	**1.12**	1.02	**0.98**	0.99	**0.98**	0.99
$+NFCI^{m}_{t,w}$	1.17	**1.16**	1.14	1.03	1.00	1.01	0.98	0.99
$+NFCI^{\theta}_{t,w}$	1.18	1.19	1.13	**1.01**	0.99	**0.99**	0.98	**0.99**

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The QR-SPF model refers to a quantile regression including SPF consensus forecasts. $NFCI_{t,w}$ adds the weekly NFCI, $NFCI^{m}_{t,w}$ denotes inclusion of three monthly lags, and $NFCI^{\theta}_{t,w}$ denotes a MIDAS specification of the NFCI. All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.9: Rolling window out-of-sample evaluation using NFCI and SPF

	week							
10% quantile score	1	3	5	7	9	11	13	15
AR-GARCH	0.42	0.41	0.39	0.39	0.35	0.35	0.36	0.36
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.44	0.42	0.36	0.37	0.42	0.37	0.30	0.33
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.41	0.38	0.35	0.35	0.33	0.34	0.33	0.34
$+x^{\mu}_{t,w} = SPF_{t,w}$	0.38	0.38	0.36	**0.29**	**0.29**	**0.29**	**0.29**	**0.29**
QR-AR	0.45	0.46	0.43	0.43	0.42	0.42	0.42	0.42
$+NFCI_{t,w}$	**0.32**	**0.33**	**0.33**	0.32	0.31	0.33	0.33	0.33
$+SPF_{t,w}$	0.42	0.42	0.40	0.34	0.31	0.31	0.31	0.31
90% quantile score	1	3	5	7	9	11	13	15
AR-GARCH	0.39	0.39	0.33	0.33	0.33	0.34	0.34	0.34
$+x^{\mu}_{t,w} = NFCI_{t,w}$	0.42	0.41	0.39	0.41	0.48	0.43	0.39	0.41
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	0.44	0.45	0.38	0.38	0.37	0.37	0.37	0.37
$+x^{\mu}_{t,w} = SPF_{t,w}$	**0.34**	**0.34**	**0.32**	**0.30**	**0.29**	**0.30**	**0.30**	**0.30**
QR-AR	0.40	0.40	0.35	0.35	0.36	0.36	0.37	0.37
$+NFCI_{t,w}$	0.42	0.42	0.36	0.35	0.35	0.36	0.37	0.38
$+SPF_{t,w}$	0.37	0.37	0.34	0.32	0.31	0.31	0.32	0.32
CRPS (equal weights)	1	3	5	7	9	11	13	15
AR-GARCH	1.23	1.21	1.13	1.13	1.12	1.12	1.13	1.13
$+x^{\mu}_{t,w} = NFCI_{t,w}$	1.29	1.27	1.16	1.18	1.30	1.21	1.12	1.16
$+x^{\sigma}_{t,w} = NFCI_{t,w}$	1.31	1.32	1.16	1.17	1.16	1.17	1.16	1.18
$+x^{\mu}_{t,w} = SPF_{t,w}$	**1.16**	**1.16**	**1.08**	**0.98**	**0.95**	**0.96**	**0.96**	**0.96**
QR-AR	1.24	1.24	1.15	1.15	1.14	1.14	1.15	1.15
$+NFCI_{t,w}$	1.22	1.25	1.17	1.17	1.16	1.17	1.17	1.17
$+SPF_{t,w}$	1.19	1.19	1.10	1.00	0.97	0.98	0.98	0.98

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. The bottom panel presents the average continuously ranked probability score (CRPS). Columns indicate the week of the quarter when the nowcast is formed. Gray shading denotes the 90% model confidence set, and bold values highlight the lowest average losses within each column. The AR-GARCH model refers to the AR(1)–GARCH(1,1) specification, while SPF-GARCH augments the conditional mean with consensus forecasts from the SPF. The QR-AR model refers to a quantile regression with an AR(1) specification. All models are estimated using a rolling window of 85 observations. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.

Table B.10: Out-of-sample evaluation using BMF-SV

	week							
10% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.36	0.35	0.35	0.28	0.29	0.29	0.29	0.29
BMF-SV	**0.33**	**0.32**	**0.32**	**0.28**	**0.26**	**0.26**	**0.26**	**0.26**
90% quantile score	1	3	5	7	9	11	13	15
SPF-GARCH	0.34	0.34	0.34	0.31	0.31	0.31	0.31	0.31
BMF-SV	**0.31**	**0.31**	**0.31**	**0.28**	**0.28**	**0.28**	**0.27**	**0.27**

Notes: This table reports average losses for the 10% and 90% quantile scores in the top and middle panels, respectively. Columns indicate the week of the quarter when the nowcast is formed. Bold values highlight the lowest average losses within each column. The SPF-GARCH model denotes an AR(1)–GARCH(1,1) specification augmented with consensus forecasts from the SPF in the conditional mean. The BMF-SV model refers to the Bayesian mixed-frequency Regression with stochastic volatility as in Carriero et al. (2022). All models are estimated using an expanding window. The evaluation period covers 1990:Q1 to 2019:Q4, yielding 120 quarterly observations.